

Additional feedforward mechanism of Parkin activation via binding of phospho-UBL and RING0 in trans

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Abstract

Loss of function Parkin mutations lead to early-onset of Parkinson's disease. Parkin is an auto-inhibited ubiquitin E3 ligase activated by phosphorylation of its ubiquitin-like (Ubl) domain and ubiquitin by PINK1. Herein, we show a competitive binding mode of the phospho-Ubl and RING2 domains on the RING0 domain, which regulates Parkin activity. We show that phosphorylated Parkin can directly complex with unmodified Parkin, leading to the activation of autoinhibited Parkin in trans. Furthermore, we show that the activator element (ACT) of Parkin is required to maintain the enzyme's kinetics, and the removal of ACT slows the enzyme catalysis. We also demonstrate that ACT can activate Parkin in trans but less efficiently than when present in the cis molecule. Furthermore, crystal structure reveals a new donor ubiquitin binding pocket that plays a crucial role in function. This study provides new molecular insights into Parkin activation, which may aid the development of Parkin activators as a therapeutic strategy against Parkinson's.

eLife assessment

This is a **valuable** manuscript describing the competitive binding between the RING2 and phosphorylated Ubl domains within Parkin involved in the regulation of Parkin activity. The evidence supporting this conclusion is **incomplete**, as it primarily relies on a single biochemical assay and does not utilize more stringent, quantitative biophysical approaches to probe this competitive binding. This work will be of interest to the research communities focused on the molecular basis of ubiquitin ligase regulation, PINK-PARKIN-regulated mitophagy, and mitochondrial quality control.

Introduction

Parkinson's disease (PD) is a neurodegenerative disorder characterized by the loss of dopaminergic neurons in the substantia nigra leading to motor defects. PD is primarily sporadic occurring mainly in older people. Mutations in several genes, such as *PARK2* (Parkin) and *PARK6* (PINK1, PTEN-induced kinase 1), cause early-onset autosomal recessive juvenile parkinsonism (ARJP). Parkin and PINK1 function together in a common mitochondrial homeostasis pathway in which damaged mitochondria are cleared by autophagy (mitophagy) ^{1,2,3-7}.

Parkin is an autoinhibited RBR family E3 ubiquitin ligase ⁸ consisting of an N-terminal ubiquitin-like (Ubl) domain followed by four Zn²⁺ binding domains RING0, RING1, inbetweenRING (IBR), and RING2 domains ⁹. Parkin is a cytosolic protein activated following mitochondrial stress, mediated by PINK1 phosphorylation of Serine 65 (Ser65) on ubiquitin, which enhances binding with Parkin and leads to the recruitment of Parkin to sites of damaged mitochondria ¹⁰⁻¹². On mitochondria, Ser65 of the Ubl domain of Parkin is phosphorylated by PINK1 ^{11,13-15}, resulting in a fully active Parkin conformation. Fully active Parkin attaches new ubiquitin molecules on mitochondrial proteins that PINK1 phosphorylates to recruit more cytoplasmic Parkin to the mitochondria, resulting in a positive feedforward amplification cycle ¹⁶. Ubiquitination of mitochondrial proteins by Parkin also leads to the recruitment of other receptors required for mitophagy ^{17,18}.

Like other RBR-family E3 ligases, Parkin binds to an E2, and ubiquitin is transferred from E2 onto the catalytic Cys431 (on RING2) of Parkin before ubiquitination of lysine residues of target substrates ^{19,20}. On Parkin, several elements are present which maintain its autoinhibited conformation. The E2 binding site on RING1 is blocked mainly by the Ubl domain, and to some extent, the short repressor (REP) element also restricts E2 binding. Furthermore, Cys431 on RING2 is occluded by the RING0 domain of Parkin, which causes inhibition of Parkin activity ^{8,21-23}. Phosphorylation of the Ubl domain results in binding with a basic patch on RING0, displacing nearby RING2 to expose Cys431 to activate Parkin ²⁴. In the structure of phospho-Parkin with RING2 removed, an activating element (ACT, 101-109), which is present in the linker region (77-140) between Ubl and RING0 domains, binds on the RING0 interface ²⁴. Mutations in the ACT were shown to affect Parkin activity negatively ²⁴, suggesting their importance in Parkin regulation. Phospho-ubiquitin (pUb) binds in a pocket between RING0 and RING1 and activates Parkin allosterically ²⁵⁻²⁷. pUb binding results in the displacement of the IBR domain and the straightening of helix-1 of the RING1 domain ²⁸. Massive domain rearrangements have been proposed in the active state to allow the transfer of donor ubiquitin (between helix-1 and IBR) from E2 (on RING1) to Cys431 (on RING2) of Parkin ^{24,28,29,30}.

Several crystal structures of Parkin were solved in the last decade using various truncations in Parkin, which revealed new insights into the conformational changes during the intricate activation process of Parkin (**Extended Data Fig. 1A** ²⁴). A few years ago, using the structure of truncated phospho-Parkin (RING2 removed) (**Extended Data Fig. 1A** ²⁴), a model of phospho-Parkin was proposed wherein RING2 would be displaced from RING0 to occupy a pocket near the IBR domain (**Extended Data Fig. 1B** ^{24,29}). However, the extent of conformational changes and domain rearrangements due to different regulatory elements of Parkin in the active state remains elusive. For example, it is unclear how and by what mechanism the displaced pUbl from RING1 would be recognized on RING0 in the cis molecule (as per the proposed model in **Extended Data Fig. 1B** ²⁴), and not in the trans molecule, especially considering the likelihood of an encounter with a trans molecule in the crowded molecular environment. Previous cellular data co-expressing WT-Parkin and mutant Parkin constructs had suggested the self-association of Parkin molecules after PINK1 activation ³¹ although an alternate explanation for these findings

would be the contribution of pUb to Parkin binding^{10–16}. Furthermore, structural studies to understand the Parkin activation mechanism in the last decade do not capture any dimerization of Parkin *in vitro*^{19–30}.

Here, using x-ray crystal structures, biophysical methods, and *in vitro* assays, we demonstrate the conformational changes in Parkin during the activation process and reveal novel insights into the Parkin activation mechanism. Our data suggest that phospho-Ubl (pUbl) domain of Parkin transiently binds with the basic patch on RING0 and competes with the RING2 domain of Parkin. In addition to the previous observation that pUbl binding results in RING2 displacement, our new data show that the presence of RING2 restricts the binding of pUbl with the Parkin core, which establishes the competitive mode of interaction between RING2 and pUbl. The crystal structure of pUbl-linker (1-140) depleted Parkin (141-465) in complex with pUb, and supporting data show that RING2 is displaced transiently during the activation process and returns to its closed state after the removal of the pUbl domain from phospho-Parkin suggesting dynamic nature of conformational changes during Parkin activation. Furthermore, we report Parkin dimerization, mediated by interactions between pUbl and basic patch on RING0 domain in trans. We also demonstrate that activated Parkin (phospho-Parkin) works as an activator of autoinhibited Parkin molecule, which suggests an additional feedforward mechanism of Parkin activation. Our data also shows new insights into the regulation mediated by the ACT of Parkin, wherein the ACT is required for maintaining the enzyme kinetics rather than any functional aspect. We also show that similar to phospho-Ubl, ACT can also work in trans; though, ACT is more efficient when present on a cis molecule. Furthermore, using x-ray crystallography and supporting experiments, we have characterized a new ubiquitin-binding site in the linker region (408-415) between the REP element and RING2, which plays a crucial role in Parkin activity. These novel insights into the Parkin activation mechanism may help accelerate the efforts to develop small-molecule activators against Parkin to treat PD.

Results

Incorporation of molecular scissors to capture intricate dynamic conformations on Parkin

Previous studies using various biophysical methods showed that after phosphorylation of the Ubl domain of Parkin, phospho-Ubl does not interact with the core of Parkin, lacking the Ubl domain^{25–27}. However, the crystal structure of RING2 removed phospho-Parkin (1-382) showed phosphorylated Ubl domain binds with the basic patch (Lys161, Arg163, Lys211) on the RING0 domain^{24,29}. RING2 shares a large surface with RING0, and the superposition of phospho-Parkin (1-382, PDB 6GLC) and WT-Parkin (PDB 5C1Z) structures show steric clashes between RING2, ACT, and pUbl (**Extended Data Fig. 2A**). Therefore, we wondered whether the RING2 domain competes with the pUbl domain and thus blocks the interaction of pUbl with RING0. The latter hypothesis would also explain why previous attempts to study pUbl interactions show weak or no interactions between pUbl and Parkin in trans.

To capture crystal structures of protein-protein complexes, researchers use fusion construct to allow the expression of two proteins in a single polypeptide chain. The fusion method increases the effective net concentration of two proteins in solution compared to mixing two proteins separately, thus stabilizing the interactions between two proteins. Earlier binding assays on Parkin failed to capture interactions in trans, and we speculated that this might be due to the lower net concentration of the domain in trans compared to the high net concentration of fused domain. We hypothesized that untethering (cleavage of peptide bond) upon protease treatment would solve the above problem and enable us to capture the binding in trans using biophysical methods. To understand the above intricate mechanism, we introduced molecular scissors (human rhinovirus type 3C (HRV 3C) and tobacco etch virus (TEV) on full-length Parkin to analyze the Ubl and RING2

domain rearrangements under native or phosphorylated states. We introduced HRV 3C (between 140th-141st residue) or TEV (382nd-383rd) sites in the loop regions of Parkin (**Extended Data Fig. 1C**) to avoid any artifacts due to perturbations in native interactions on protein.

First, we tested the ubiquitination activity of the Parkin construct with HRV 3C and TEV sites to ensure that the inclusion of these sites did not affect the protein folding or function, which is confirmed by the similar activity of this construct as of the native Parkin construct (**Extended Data Fig. 2B**). Furthermore, we noticed that Ubl-linker (1-140) co-elute with R0RBR (141-465) in native Parkin after treatment with 3C protease, suggesting a stronger interaction between Ubl and the Parkin core (**Extended Data Fig. 2C**). However, when phosphorylated Parkin was treated with 3C, pUbl-linker (1-140) did not form a complex with R0RBR (141-465), suggesting a poor/no interaction between phospho-Ubl with the core of Parkin (**Extended Data Fig. 2C**). Furthermore, when we treated native Parkin with TEV, RING2 (383-465) co-elute with Parkin (1-382), suggesting a stronger interaction between RING2 and the Parkin core in the native Parkin (**Extended Data Fig. 2D**). However, when we treated phosphorylated Parkin with TEV, RING2 (383-465) elute separately from the Parkin (1-382), suggesting that phosphorylation of the Ubl domain results in the displacement of the RING2 domain (**Extended Data Fig. 2D**). All the above data confirmed that inclusion of molecular scissors on Parkin constructs do not affect Parkin folding and the previous observations, that phosphorylation of Ubl weakens its interaction with Parkin core and leads to displacement of RING2, are confirmed using our assay. Also, respective proteases only cleaved (untethered) the peptide bond without affecting the native interactions between Parkin domains.

The phospho-Ubl domain and RING2 domain have a competitive mode of binding on the RING0 domain

We wanted to test whether RING2 and pUbl affect the binding of each other on Parkin, which would suggest a competitive binding mode between pUbl and RING2 on the RING0 domain, or as suggested previously, pUbl negatively affects RING2 binding and lead to RING2 displacement permanently. To test the competitive mode of binding between pUbl and RING2 on RING0, and thus affecting the binding of each other, we performed the SEC assay after sequential treatment with HRV 3C and TEV on Parkin construct with both protease sites as shown in **Extended Data Fig. 1C**. Interestingly, pUbl-linker (1-140) co-elute with Parkin core (141-382) upon 3C treatment on fractions that were collected after TEV treatment on phospho-Parkin which lead to displacement of RING2 (383-465) (**Fig. 1A**). Similarly, RING2 (383-465) co-elute with Parkin core (141-382) upon TEV treatment on fractions that were collected after 3C treatment on phospho-Parkin, which led to displacement of pUbl-linker (1-140) (**Fig. 1B**). This data confirms that pUbl and RING2 competitively bind on RING0. The binding of one negatively affects the binding of the other, unlike previous observations, which only showed phosphorylation of Ubl leading to RING2 displacement.

Furthermore, we wanted to test whether RING2 is permanently displaced from RING0 after the pUbl binding on the RING0 domain. To test this hypothesis, we used phosphorylated Parkin, which had both TEV and 3C sites. We crystallized the phospho-Parkin complex with phospho-ubiquitin after treatment with 3C protease, which resulted in the washing off of pUbl-linker (1-140) from Parkin (141-465). The overall structure of pUbl-linker (1-140) depleted Parkin (141-465) in complex with pUb was determined at 3.3 Å (**Extended Data Table 1**), and showed similar conformation as seen in previously solved Parkin structures in their autoinhibited state (**Fig. 1C**). The crystal structure shows RING2 bound with RING0, which confirms that RING2 is transiently displaced from the RING0 domain and returns to its original position after washing off pUbl-linker (**Fig. 1C**), further confirming our SEC data (**Fig. 1B**). The crystal structure also revealed that the REP element is bound with the RING1, similar to the Parkin structures captured in the autoinhibited state (**Fig. 1C**). The phospho-ubiquitin was seen bound in the basic patch between RING0 and RING1 and resulted in similar conformational changes in IBR and helix between RING1-IBR domains (**Fig. 1C**). In the crystal structure, two molecules of Parkin bound with pUb

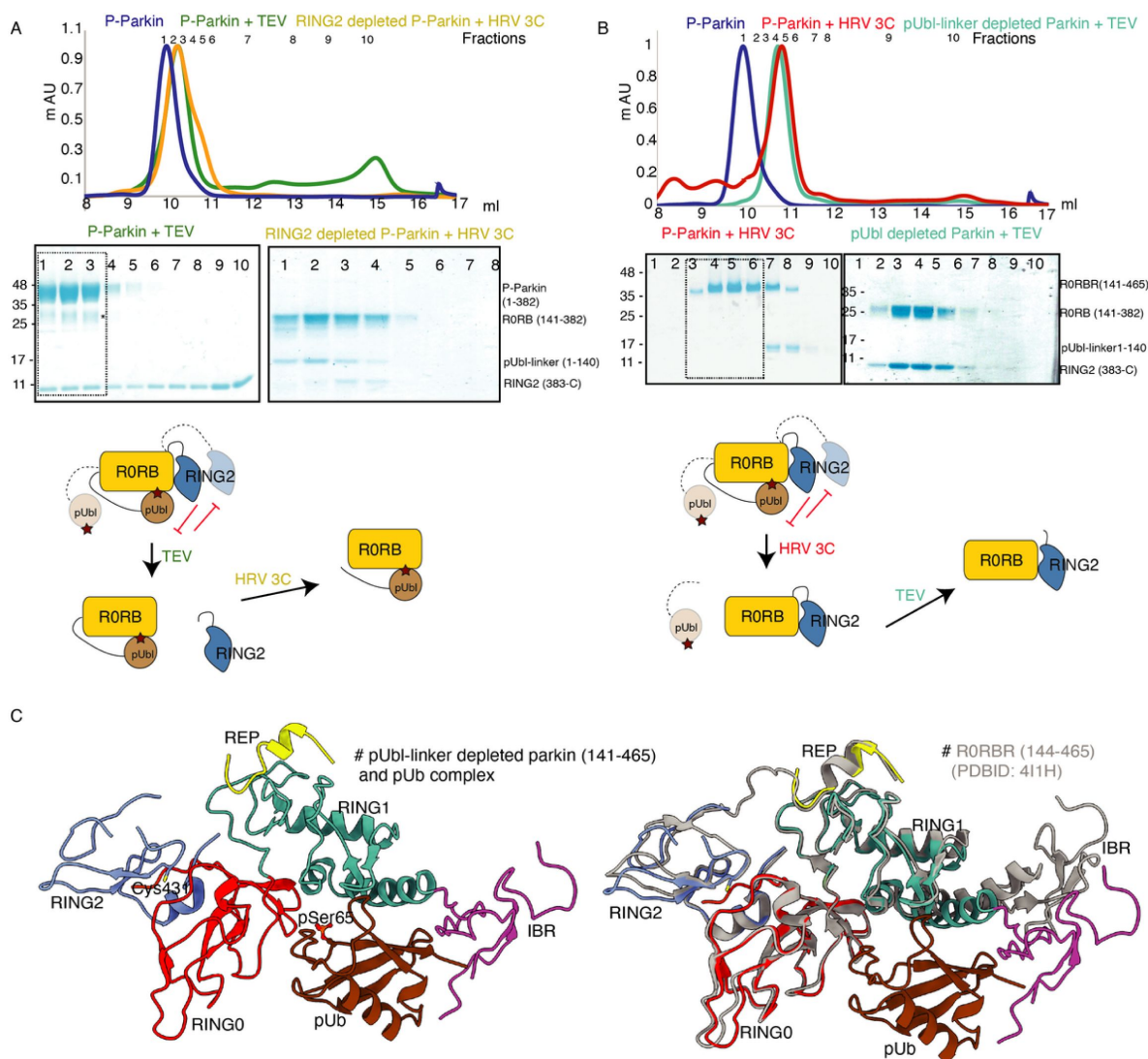


Figure 1

Characterization of pUbl and RING2 interactions on RING0

A. SEC assay showing depletion of RING2 (383-465) from phospho-Parkin stabilize pUbl-linker (1-140) binding with Parkin (141-382) after treatment with 3C protease. Fractions that were pooled for subsequent proteolysis are highlighted in the box.

B. SEC assay showing depletion of pUbl-linker (1-140) from phospho-Parkin stabilize RING2 (383-465) binding with Parkin (R0RB, 141-382) after treatment with TEV protease. Fractions that were pooled for subsequent proteolysis are highlighted in the box.

C. Crystal structure of pUbl-linker (1-140) depleted Parkin (141-465) in complex with pUb (brown). Different domains of Parkin are colored, as shown in the left panel. Catalytic Cys431 is highlighted. pUbl-linker (1-140) depleted Parkin (141-465) complex structure (colored as in the left panel) is superimposed with R0RBR structure (PDB: 4I1H, grey) in the right panel.

were seen; however, in one of the Parkin molecules, no density was observed in the IBR region (**Extended Data Fig. 3**). Overall, this data suggest that pUbl and RING2 exist in a dynamic state in phospho-Parkin (pUbl binding $\leftrightarrow$ RING2 open $\leftrightarrow$ pUbl displaced $\leftrightarrow$ RING2 closed) and compete for binding on RING0, unlike previous studies suggesting that only the open conformation of Parkin is mediated by RING2 displacement by pUbl.

Interestingly, similar to phospho-Parkin (**Extended Data Fig. 2C**), pUbl-linker (1-140) remained flexible in phospho-Parkin (Lys211Asn) and eluted separately from Parkin core (141-465) on SEC (**Extended Data Fig. 4A**), suggesting binding with the basic patch on RING0 domain may not be the driving force for pUbl displacement. Further, to confirm that displacement of the RING2 domain is mediated by pUbl binding in the basic patch (Lys161, Arg163, and Lys211) on the RING0 domain, we tested the RING2 displacement using Lys211Asn mutant of Parkin. Mutation of Lys211Asn resulted in stabilization of RING2 (383-465) domain on phospho-Parkin (1-382) upon TEV treatment, and the two fragments co-eluted on SEC (**Extended Data Fig. 4A**). Although pUbl was displaced in phospho-Parkin (Lys211Asn), Lys211Asn mutation reduced Parkin activity drastically (**Extended Data Fig. 4B**), suggesting RING2 displacement, not Ubl displacement, is a major cause of Parkin activation. Also, we noticed a basal level of Parkin activity in the lanes without any activator, which was reduced in Lys211Asn Parkin (**Extended Data Fig. 4B**). To understand the conformational changes upon mutation in the basic patch on RING0, we also crystallized phospho-Parkin (Arg163Asp, Lys211Asn, HRV 3C site at 141st position) in complex with phospho-ubiquitin after treatment with 3C protease, which resulted in washing off of pUbl-linker (1-140) from Parkin core (141-465). This complex resulted in better crystals diffracting up to 2.35 Å. The overall structure of the pUbl-linker (1-140) depleted Parkin (141-465, Arg163Asp, Lys211Asn) in complex with pUb is similar to the autoinhibited structure wherein RING2 is bound on RING0 and REP element is bound on RING1 (**Extended Data Fig. 4C**).

Untethering of the linker connecting IBR and RING2 allows pUbl binding in trans

We wondered whether the competing nature of binding between pUbl and RING2 could be the reason behind no interaction seen previously between pUbl and Parkin (lacking Ubl domain) in trans^{19,30}. To test this, we used phospho-Parkin (Lys211Asn), which would not allow the binding of pUbl in the RING0 pocket of the same molecule, and tested its interaction with Δ Ubl-Parkin. However, no complex formation was seen on SEC (**Fig. 2A**). SEC assay was validated using ITC assay which did not show any detectable interaction between phospho-Parkin (Lys211Asn) and Δ Ubl-Parkin (**Fig. 2A**). This trans experiment is consistent with our cis experiment (**Fig. 1**). As our data suggested that the fused domain outcompete the untethered domain (**Fig. 1**), we wondered whether the same could be the reason behind the lack of detectable binding in trans. To test this, we used acceptor Parkin with an untethered linker (TEV site between 382nd - 383rd), which overcomes the problem of higher net concentration of the fused competing RING2 domain. Acceptor Δ Ubl-Parkin (TEV site between 382nd - 383rd) was treated with TEV, and TEV was removed using an affinity column followed by SEC. SEC showed co-elution of Δ Ubl-Parkin (77-382) and RING2 (383-465), confirming that TEV cleaved (untethered) the peptide bond (connecting IBR and REP-RING2) without affecting the native interactions between Δ Ubl-Parkin (77-382) and RING2 (383-465) (**Fig. 2B**). Incubation of phospho-Parkin (Lys211Asn) with untethered Δ Ubl-Parkin led to a stable trans complex on the SEC, showing co-elution of phospho-Parkin (Lys211Asn) and Δ Ubl-Parkin (77-382), leading to the displacement of RING2 (383-465) from Δ Ubl-Parkin (**Fig. 2B**). The ITC assay showed a strong affinity ($K_d = 1.3 \pm 0.2 \mu\text{M}$) between phospho-Parkin (Lys211Asn) and untethered Δ Ubl-Parkin (**Fig. 2B**), which further supports SEC assay. We noticed protein precipitation in the cell during ITC measurements, which explains the poor stoichiometry due to protein destabilization after pUbl binding and RING2 displacement. A significant decrease of 3 °C in the melting temperature of phospho-Parkin²⁴ is consistent with the above observation.

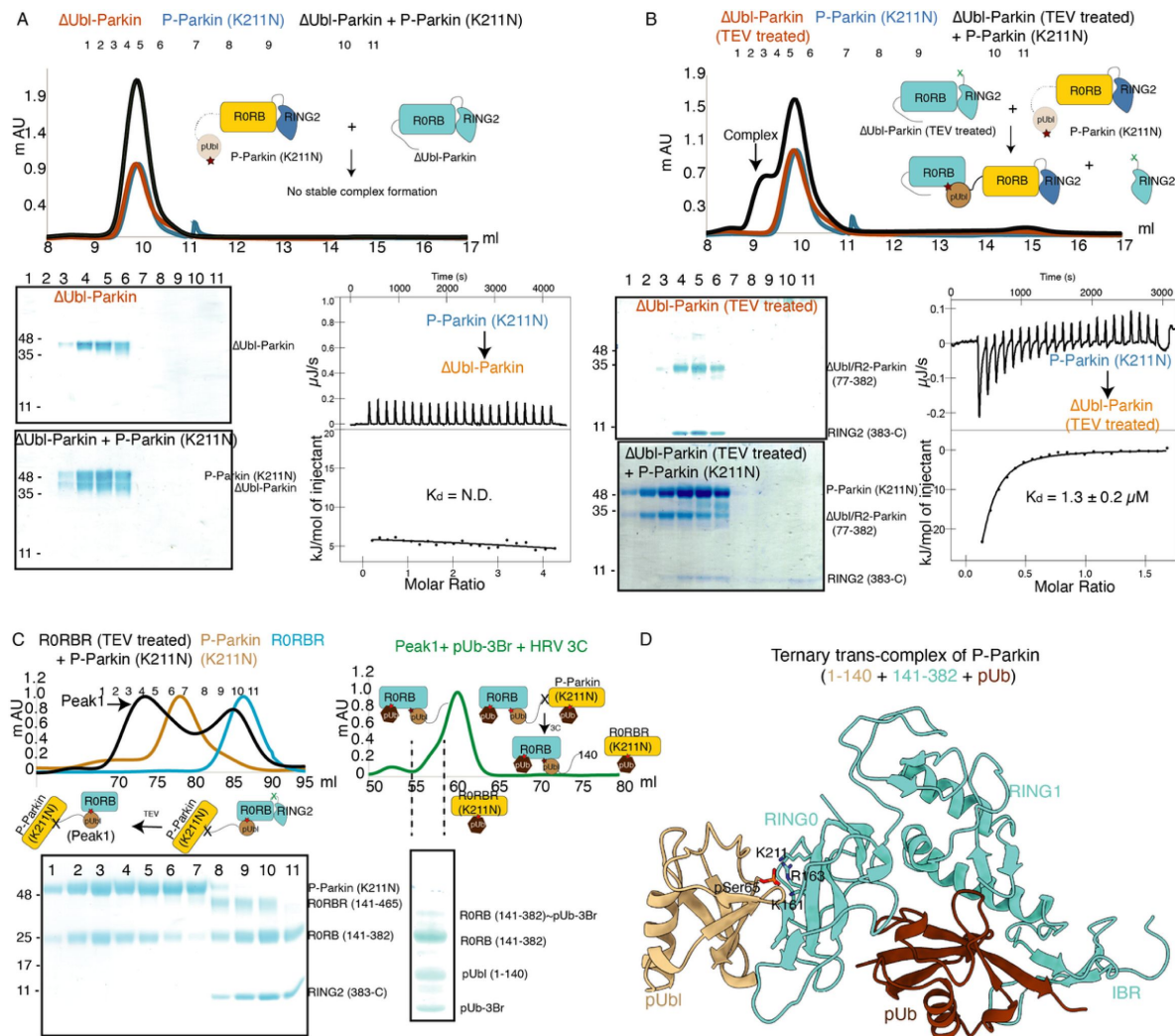


Figure 2

Untethering of the linker between IBR-RING2 allows Parkin and phospho-ubl interaction in trans

A. Binding assay between phospho-Parkin (Lys211Asn) and ΔUbl-Parkin. A colored key for each trace is provided. Coomassie-stained gels of indicated peaks are shown in the lower panel. A schematic representation is used to explain SEC data. In the lower panel, the Isothermal Titration Calorimetry assay between phospho-Parkin (Lys211Asn) and ΔUbl-Parkin is shown. N.D. stands for not determined.

B. Binding assay between phospho-Parkin (Lys211Asn) and ΔUbl-Parkin (untethered RING2 by TEV treatment). A colored key for each trace is provided. Coomassie-stained gels of indicated peaks are shown in the lower panel. A schematic representation is used to explain SEC data. In the lower panel, the Isothermal Titration Calorimetry assay between phospho-Parkin (Lys211Asn) and ΔUbl-Parkin (untethered RING2 by TEV treatment) is shown. The dissociation constant (K_d) is shown.

C. SEC assay to test binding between R0RBR (untethered RING2 by TEV treatment) and phospho-Parkin (Lys211Asn), and displacement of RING2 (383-465) from R0RBR, the left panel. The peak1 (black) containing R0RBR (141-382) and phospho-Parkin (Lys211Asn) complex was incubated with pUb, followed by HRV 3C, to purify ternary trans-complex of phospho-Parkin (1-140 + 141-382 + pUb) on SEC, the right panel. The concentrated fractions from the shoulder (highlighted with a dashed line) of the peak in the right panel were loaded on SDS PAGE to confirm complex formation.

D. Crystal structure of the trans-complex of phospho-Parkin with pUb (brown) showing phospho-Ubl domain (wheat) bound with RING0 (cyan) domain of Parkin (cyan).

Further, to confirm that untethering does not affect the native interactions between RING2 and RING0 domains, we purified and determined the structure of TEV-treated R0RBR Parkin (141-465, TEV site between 382nd - 383rd, **Extended Data Fig. 5A** [↗](#)). Co-elution of R0RB (141-382) and RING2 (383-465) fragments on SEC (**Extended Data Fig. 5B** [↗](#)) and crystal structure showing intact native interactions between RING2 and RING0 (**Extended Data Fig. 5C** [↗](#)) further ruled out any possibility of artifact.

Additionally, we wanted to understand the molecular details of the complex seen in **Figure 2B** [↗](#). We used FL-Parkin (Lys211Asn, HRV 3C site between 140th and 141st) as a donor of pUbl-linker (1-140), and R0RBR Parkin (TEV site between 382nd-383rd) as an acceptor of pUbl-linker. Phospho-Parkin (Lys211Asn) formed a stable complex with R0RBR protein with untethered RING2, and RING2 (383-465) was removed from R0RBR (**Fig. 2C** [↗](#)). The fractions containing the complex of phospho-Parkin (Lys211Asn) and R0RB (141-382) upon treatment with 3C followed by incubation with pUb showed co-elution of components of the ternary trans-complex (R0RB (141-382), pUbl-linker (1-140), and pUb) on SEC (**Fig. 2C** [↗](#)). The crystal structure of the ternary trans-complex of phospho-Parkin (pUbl-linker (1-140) + R0RB (141-382) + pUb) solved at 1.92 Å (**Extended Data Table 1** [↗](#)) further confirmed trans-complex formation between Parkin molecules (**Fig. 2D** [↗](#), **Extended Data Fig. 6** [↗](#)). In the crystal structure, the pUbl domain from the donor molecule binds in the basic patch of RING0 of the acceptor molecule (**Fig. 2D** [↗](#)) in trans, similar to what is previously seen in the phospho-Parkin (1-382) structure with fused pUbl domain and untethered/truncated RING2 in cis molecule [24,29](#) [↗](#). Interestingly, the linker between pUbl and RING0 remained disordered in all the structures [24,29](#) [↗](#). Therefore, it would be difficult to say whether, in the previous cis structure, the pUbl bound to RING0 was from the same molecule or different molecules. Also, the fusion of pUbl with RING0 and untethering/truncation of RING2 as in the previous structures [24,29](#) [↗](#) may favor pUbl binding with RING0 in cis. Our data establish that keeping pUbl and RING2 untethered from their binding partner RING0, thus reducing the artifact due to the higher net concentration of the fused domain with RING0, is ideal for measuring trans interactions using biophysical methods.

Phospho-Parkin activates native Parkin in trans

As the pUbl domain remains dynamic in both native phospho-Parkin and phospho-Parkin Lys211Asn (**Extended Data Fig 2C** [↗](#), **Extended Data Fig 4A** [↗](#)), we wondered whether a trans-complex is formed using native phospho-Parkin. To test this, we used native phospho-Parkin (1-465) as a pUbl donor and ΔUbl-Parkin (77-465, TEV treated) as a pUbl acceptor on RING0. Interestingly, phospho-Parkin forms a stable complex with ΔUbl-Parkin (77-382) and results in the removal of RING2 (383-465) from ΔUbl-Parkin (**Fig. 3A** [↗](#), **Extended Data Fig. 7A** [↗](#)). We further tested the binding of phospho-Parkin (1-465) with WT-Parkin (1-465, TEV treated) as a pUbl acceptor. Similar to ΔUbl-Parkin, phospho-Parkin forms a stable complex with WT-Parkin (1-382) and results in the removal of RING2 (383-465) from WT-Parkin (**Fig. 3B** [↗](#), **Extended Data Fig. 7B** [↗](#)). However, mutation in the basic patch of Parkin (1-465, Lys211Asn, TEV treated) perturbed the complex formation with phospho-Parkin (**Fig. 3B** [↗](#), **Extended Data Fig. 7C** [↗](#)) confirming that interactions between pUbl and the basic patch on the RING0 domain form trans-complex. We also confirmed complex formation using SEC-MALS (size-exclusion chromatography coupled with multi-angle light scattering) to rule out any ambiguity. MALS analysis further confirms complex (Observed M. W. = 94 ± 3 Kda) formation when phospho-Parkin (Observed M. W. = 53 ± 2 Kda) was mixed with TEV-treated WT-Parkin (Observed M. W. = 52 ± 3 Kda) (**Fig 3C** [↗](#), **Extended Data Fig. 7B** [↗](#)).

As our binding experiments suggested that phosphorylated Parkin can bind with unphosphorylated Parkin, we wanted to check whether phosphorylated Parkin can activate unphosphorylated native Parkin. To test phospho-Parkin mediated native Parkin activation in trans, we used a catalytic-inactive version of phospho-Parkin with mutations in both the E2 binding site (T270R) and catalytic site (C431A). Interestingly, we observed that WT-Parkin ubiquitination/autoubiquitination activity is increased with increasing concentrations of phospho-

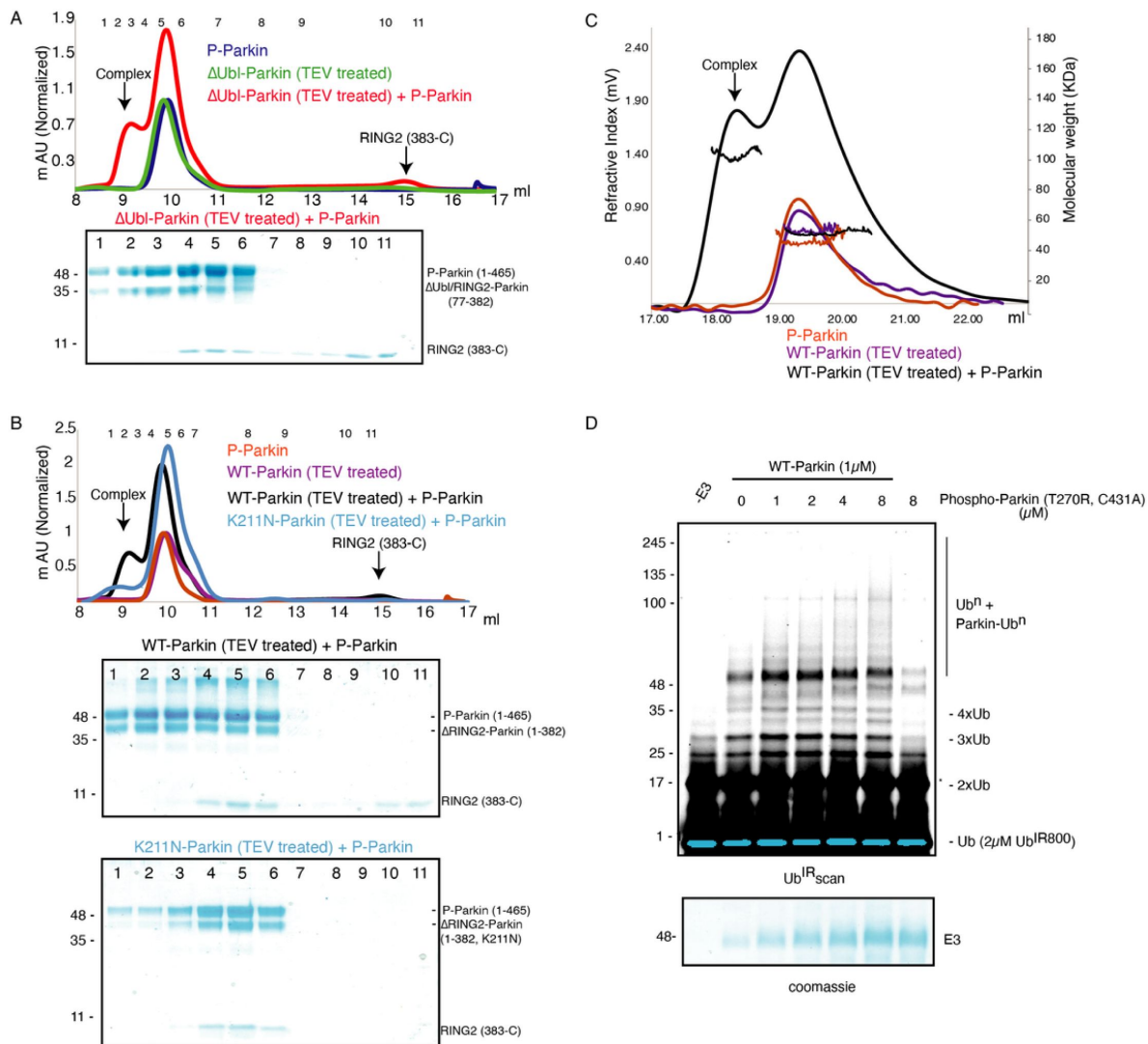


Figure 3

Parkin dimerization and trans-activation of native Parkin are mediated by phosphorylation of the Ubl domain of Parkin

A. SEC assay between phospho-Parkin and untethered Δ Ubl-Parkin. A colored key for each trace is provided. Coomassie-stained gels of indicated peaks are shown in the lower panel. TEV protein contamination is indicated (*).

B. SEC assay between phospho-Parkin and untethered WT-Parkin (upper panel) or untethered Lys211N-Parkin (lower panel). TEV protein contamination is indicated (*).

C. SEC-MALS assay to confirm the complex formation between untethered WT-Parkin and native phospho-Parkin.

D. Ubiquitination assays to check the WT-Parkin activation (right panel) with increasing concentrations of phospho-Parkin (T270R, C431A). A non-specific, ATP-independent band is indicated (*). The lower panel shows a Coomassie-stained loading control.

Parkin (T270R, C431A) (**Fig. 3D**). Although, we were not expecting activation of WT-Parkin by phospho-Parkin as Ubl of WT-Parkin would block the E2 binding site on RING1 in WT-Parkin, activation of WT-Parkin with phospho-Parkin (T270R, C431A) suggested that a significant inhibition on Parkin is mediated by RING0 blocking RING2, which is released upon binding with pUbl. Further, we wondered whether pUbl could also enhance Parkin phosphorylation similar to what is seen previously with pUb³². To test this, we checked Parkin phosphorylation by PINK1 in the presence of pUbl or pUb. However, unlike pUb, pUbl did not affect the Parkin phosphorylation by PINK1 (**Extended Data Fig. 7D**), further confirming that pUbl and pUb binding lead to unique conformational changes in Parkin. Overall, this data shows pUbl domain-mediated dimerization of Parkin molecules leading to Parkin activation in trans.

Assessment of Parkin activation in cells

It has previously been reported that pUb may interact with the RING0 domain of Parkin and that loss of this interaction underlies loss of Parkin recruitment to the mitochondria in cells expressing Lys211Asn Parkin³⁹. However, we recently showed that pUb does not bind in the RING0 pocket and specifically bind with the RING1 pocket³³ that contrasts with phospho-Ubl binding in the RING0 pocket and displacing RING2 in trans (**Fig. 3**). Biophysical assays also reveal that unlike the tight binding of pUb in the RING1, pUbl binding in the RING0 pocket is very transient in nature. Furthermore, Lys211Asn mutation in the RING0 pocket results in loss of Parkin activity by both loss of pUbl-mediated interactions (**Fig 3**) and by Asn211-driven conformational changes leading to loss of Parkin activity independent of pUb binding³³. This loss of Parkin activity would lead to a reduced amount of pUb, resulting in loss of Parkin recruitment to mitochondria. Therefore, we decided to test for any activity-independent Parkin recruitment to impaired mitochondria using a Parkin translocation assay in HeLa cells^{10,14,16,31}. Consistent with previous studies,^{10,14,16,31} full-length wild-type but not catalytic-inactive Cys431Phe GFP-Parkin is recruited to mitochondria following carbonyl cyanide m-chlorophenyl hydrazone (CCCP) treatment (**Extended Data Fig. 8A-B**). Similarly, we did not observe recruitment of GFP-Parkin Cys431Phe RING1 (His302Ala) or RING0 (Lys211Asn) mutants when expressed alone (**Extended Data Fig. 8A-B**).

However, we observed that co-expression of mCherry-tagged-Parkin WT with GFP-Parkin Cys431Phe enables GFP-Parkin Cys431Phe recruitment to the mitochondria, similar to a previous study³¹ (**Fig. 4A, D**). Under these assay conditions, we strikingly observed that mutation of the pUb binding pocket in the RING1 completely abolished recruitment of the double mutant GFP-Parkin Cys431Phe/His302Ala to the mitochondria, when co-expressed with mCherry-tagged-Parkin WT (**Fig. 4B, D**). This excludes a significant role for the RING0 pocket in pUb binding in the context of full length parkin expressed in cells following mitochondrial damage (**Fig. 4B, D**). In line with this, mutation of the RING0 binding pocket produced a moderate defect in recruitment of the double mutant GFP-Parkin Cys431Phe, Lys211Asn to the mitochondria when co-expressed with mCherry-tagged-Parkin WT (**Fig. 4C, D**), suggesting that the transient interaction between pUbl and RING0 of Parkin in trans acts in concert with pUb binding to RING1 pocket for optimal Parkin recruitment to sites of mitochondrial damage (**Fig. 4C, D**). Under all transfection conditions, we did not observe a significant difference in mCherry-tagged Parkin WT (**Extended Fig. 8C**). Furthermore, co-expression of GFP-Parkin Cys431Phe or GFP-Parkin Cys431Phe/Lys211Asn or GFP-Parkin Cys431Phe/His302Ala with the non-phosphorylatable mCherry-tagged-Parkin Ser65Ala failed to rescue recruitment to the mitochondria (**Fig. 4A, B, C, D**). These findings are in line with our biophysical data and highlight the importance of phospho-Ubl domain-mediated interactions in Parkin recruitment to the mitochondria.

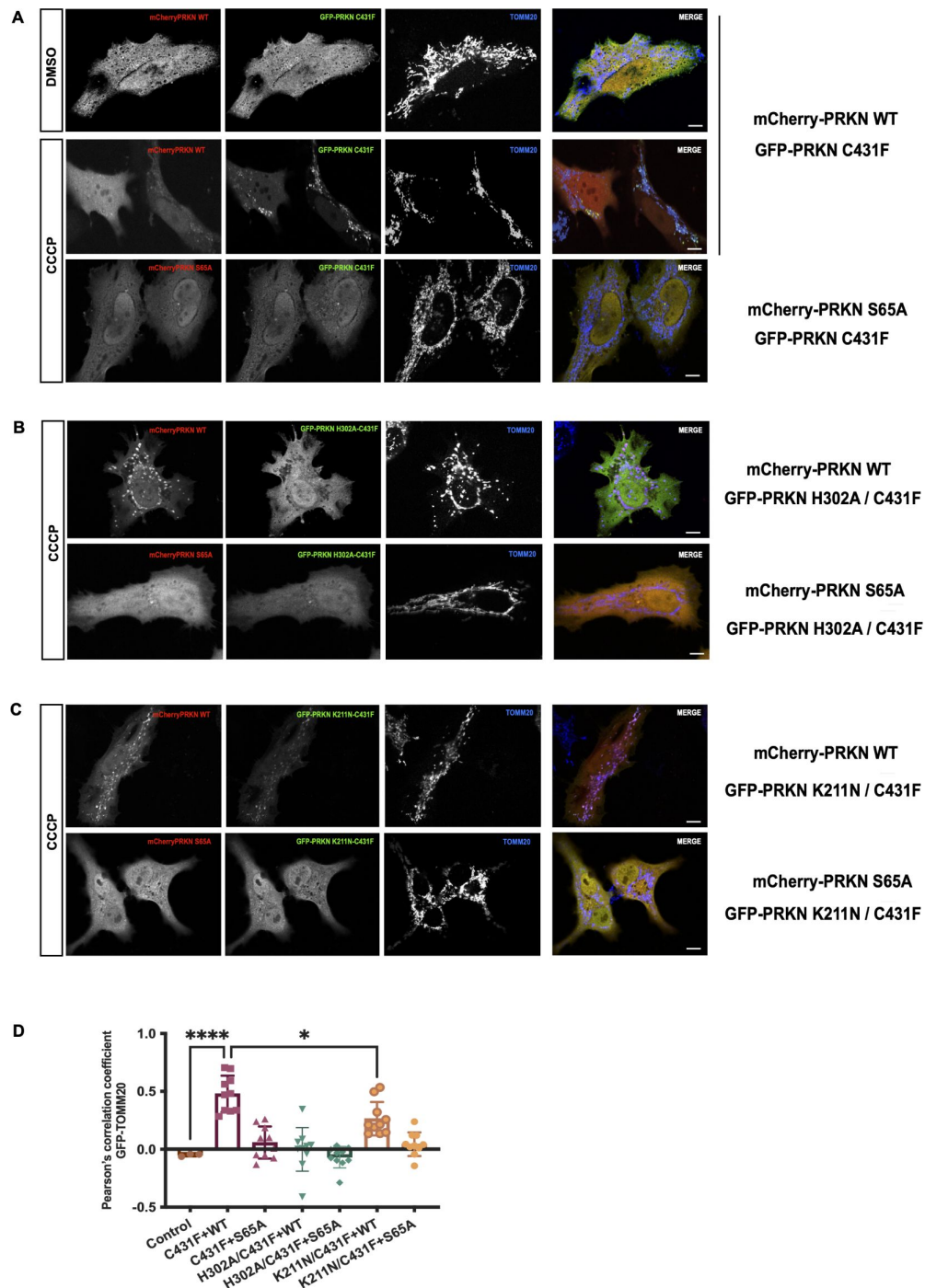


Figure 4

Analysis of Parkin mutant recruitment to mitochondria in HeLa cells

A. Immunofluorescence of HeLa cells co-transfected with either wild-type (WT) mCherry-Parkin or mCherry-Parkin S65A and GFP-Parkin C431F or, B. GFP-Parkin H302A-C431F and, C. GFP-Parkin K211N-C431F. Cells were treated for 1 h with 10 μ M CCCP, DMSO was used as control. Mitochondria were labelled with anti-TOMM20 antibody (blue). Scale bars, 10 μ m. D. Quantification of GFP-Parkin (WT and mutants) on mitochondria. The colocalization of GFP-Parkin (WT and mutants) with TOMM20 (mitochondria) was assessed using Pearson's correlation coefficient. Errors are represented as S.D. and statistical differences in Pearson's correlation coefficient were assessed by one-way ANOVA and Tukey's multiple comparisons post-test. Statistical significance is indicated as follows: *, $p < 0.05$; ****, $p < 0.0001$.

ACT improves Parkin enzyme kinetics

A previous study identified a small region (101-109) in the linker between Ubl and RING0 as an activator element (ACT) required for Parkin activity²⁴. To further explore the role of the ACT, we tested whether the omission of ACT affects the binding of Parkin with the charged state of E2 (E2~Ub). We observed that phospho-Parkin in a complex with pUb binds tightly with E2~Ub and co-elutes on SEC (**Fig. 5A**). Interestingly, deletion of the ACT did not affect the complex formation with E2~Ub, as phospho-Parkin (Δ ACT) in complex with pUb co-elutes with E2~Ub on SEC (**Fig. 5A**). As the displacement of RING2 is a crucial process during Parkin activation, we tested whether the removal of the ACT affects the displacement of the RING2 domain using our TEV-based SEC assay. We observed that phospho-Parkin (Δ 101-109/ACT) after treatment with TEV results in a shift where RING2 (383-465) is displaced from the Parkin core (1-382, Δ ACT), resulting in the elution of two fragments of Parkin separately on SEC (**Fig. 5B**). As the deletion of ACT did not show any functional defect in Parkin, we hypothesized that the presence of ACT at the interface of RING0 and RING2 might affect the dynamic nature of RING2 thereby regulating the enzyme kinetics. To test this hypothesis, we compared the kinetics of phospho-Parkin (Δ ACT) ubiquitination activity over different time points. We observed that the deletion of ACT slows the kinetics of Parkin activity, doubling the time for phospho-Parkin (Δ ACT) to reach a similar level of activity as un-modified phospho-Parkin (**Fig. 5C**).

ACT is more efficient in cis than in trans

We next examined the interactions between ACT and Parkin in our ternary trans-complex of phospho-Parkin (1-140 + 141-382 + pUb) structure solved at a similar resolution and in the same space group as the previous solved structure of phospho-Parkin (1-382) in complex with pUb²⁴. However, we do not see any density of the ACT region in the ternary trans-complex of phospho-Parkin (1-140 + 141-382 + pUb) structure (**Fig. 6A**, **Extended Data Fig. 9A**), unlike the phospho-Parkin (1-382) in complex with pUb structure where the ACT region was clearly shown to occupy the hydrophobic pocket on RING0 (**Fig. 6A**). Interestingly, we observed that in the ternary trans-complex of phospho-Parkin structure, Lys48 of the pUbl domain occupies the same pocket that Arg104 of the ACT region occupied in the phospho-Parkin in complex with pUb structure (**Fig. 6A**, **Extended Data Fig. 9A**). Also, the side-chain of Lys48 of the pUbl domain was disordered in the previous structure of phospho-Parkin (1-382) in complex with pUb (**Fig. 6A**).

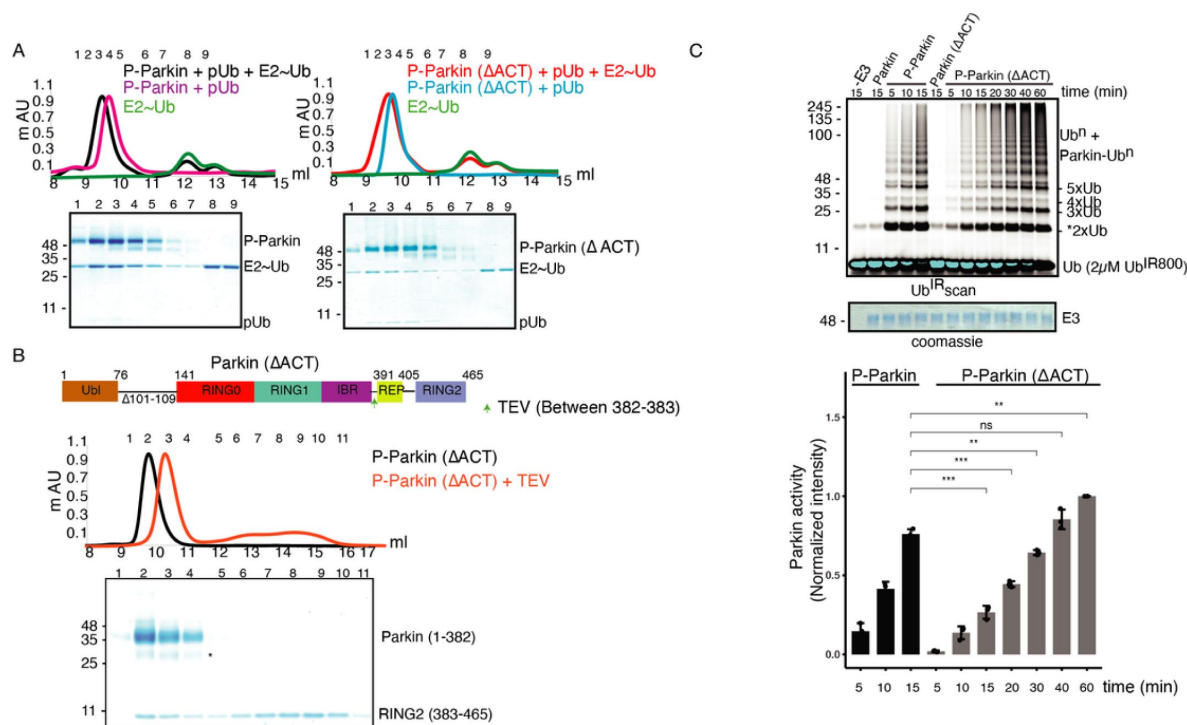


Figure 5

ACT plays a key role in enzyme kinetics

A. Size-exclusion chromatography (SEC) assay to test the binding of E2~Ub with phospho-Parkin (left panel) or phospho-Parkin (ΔACT) (right panel). Assays were done using Parkin in complex with pUb. A colored key for each trace is provided. Coomassie-stained gels of indicated peaks are shown in the lower panel.

B. Size-exclusion chromatography (SEC) assay to check displacement of the RING2 domain after phosphorylation of Parkin (ΔACT). The upper panel shows a schematic representation of the Parkin (ΔACT) construct used for the RING2 displacement assay. Conformational changes in Parkin, as suggested by the SEC experiment, are shown schematically.

C. Ubiquitination assay to check the effect of ACT deletion on Parkin activity, phospho-Parkin (ΔACT) activity shows higher activity in the lanes where reactions were incubated at longer time points as indicated. A non-specific, ATP-independent band is indicated (*). The middle panel shows a Coomassie-stained loading control. In the lower panel, the bar graph shows the integrated intensities of ubiquitin levels from three independent experiments (mean ± s.e.m.). Statistical significance was determined using pair-wise student's t-test (**P < 0.01, ***P < 0.001, ns-nonsignificant).

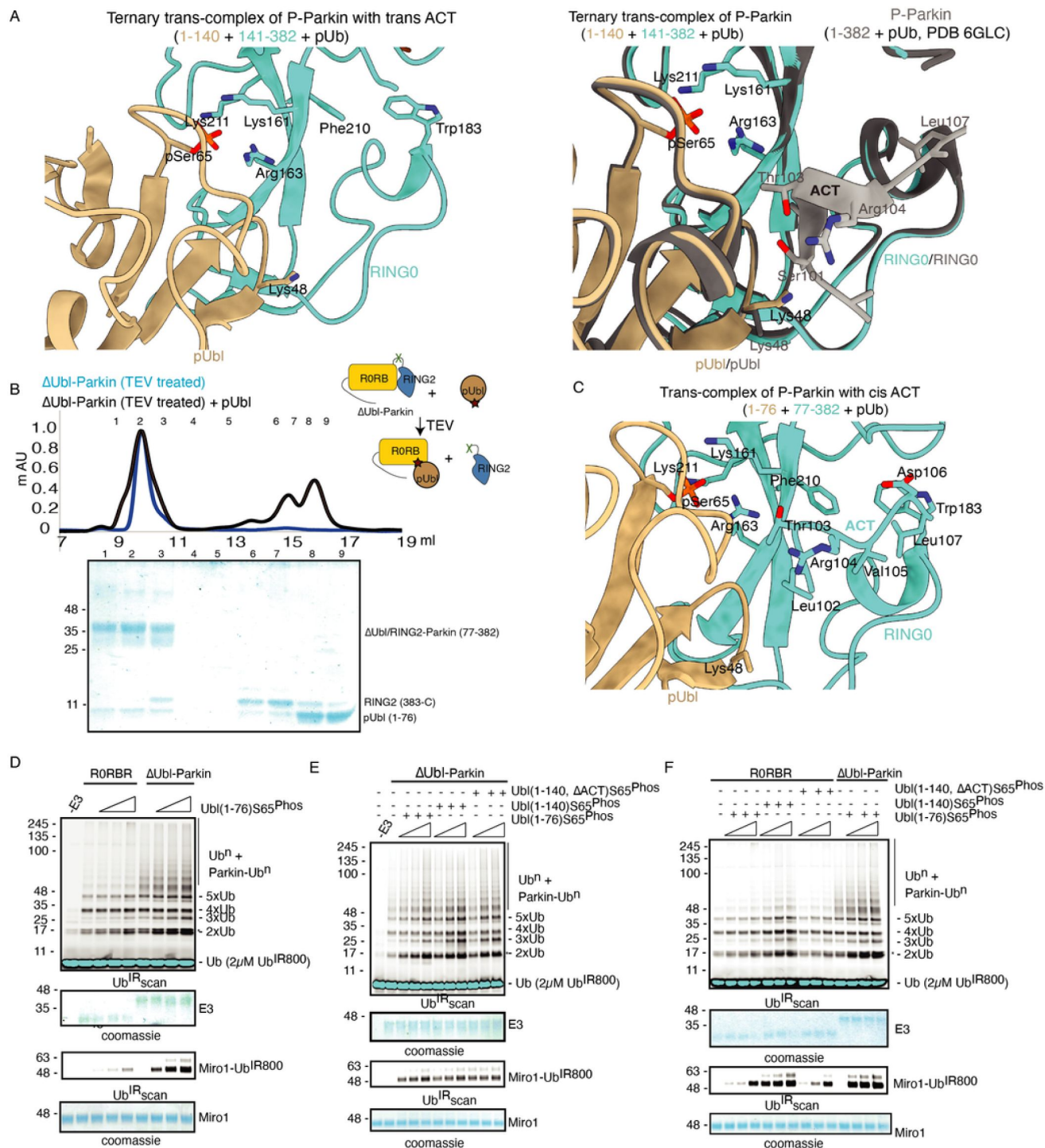


Figure 6

ACT is more efficient in cis

A. Crystal structure of ternary trans-complex of phospho-Parkin with pUb (1-140 + 141-382 + pUb), left panel. pUbl (wheat) of 1-140 fragment of Parkin and RING0 (cyan) of 141-382 fragment of Parkin are shown. The right panel shows superimposed structures of ternary trans-complex of phospho-Parkin with pUb, colored as the left panel, and the phospho-Parkin complex with pUb (PDB 6GLC) is shown in grey.

B. SEC assay to check the binding between Δ Ubl-Parkin (untethered RING2 by TEV) and phospho-Ubl (1-76). A colored key for each trace is provided. Coomassie-stained gels of indicated peaks are shown in the lower panel.

C. Crystal structure of trans-complex of phospho-Parkin with cis ACT (1-76+ 77-382 + pUb) shows ACT (cyan) present in the pocket on RING0 (Cyan) and pUbl (wheat) in the vicinity.

D. Comparison of R0RBR and Δ Ubl-Parkin activation using the increasing concentrations of pUbl (1-76). A non-specific, ATP-independent band is indicated (*). The middle panel shows a Coomassie-stained loading control. The lower panel shows Miro1 ubiquitination for the respective proteins shown in the upper lane. Coomassie-stained gel showing Miro1 is used as the loading control of substrate ubiquitination assay.

E. Ubiquitination assay of Δ Ubl-Parkin with increasing concentrations of pUbl (1-76), pUbl-linker (1-140), pUbl-linker- Δ ACT (1-140, Δ 101-109). A non-specific, ATP-independent band is indicated (*). The middle panel shows a Coomassie-stained loading control. The lower panel shows Miro1 ubiquitination for the respective proteins shown in the upper lane. Coomassie-stained gel showing Miro1 is used as the loading control of substrate ubiquitination assay.

F. Comparison of R0RBR activation using the increasing concentrations of pUbl (1-76), pUbl-linker (1-140), pUbl-linker- Δ ACT (1-140, Δ 101-109). Δ Ubl-Parkin activation reactions using increasing concentrations of pUbl (1-76) are used as control. A non-specific, ATP-independent band is indicated (*). The middle panel shows a Coomassie-stained loading control. The lower panel shows Miro1 ubiquitination for the respective proteins shown in the upper lane. Coomassie-stained gel showing Miro1 is used as the loading control of substrate ubiquitination assay.

We wondered whether the lack of density in the ACT region was due to the preference of ACT to remain associated with the cis molecule rather than to be complemented by the trans molecule. To test this hypothesis, we determined the crystal structure of the ternary trans-complex of phospho-Parkin with cis ACT using phospho-Ubl (1-76) and Δ Ubl-Parkin (77-465, TEV site between 382nd - 383rd). pUbl formed a stable complex with Δ Ubl-Parkin (77-382) after TEV treatment and resulted in the displacement of RING2 (383-465) (**Fig. 6B**). Fractions containing trans-complex of phospho-Parkin with cis ACT (1-76 + 77-382) were mixed with pUb to get the crystals of the ternary complex. The ternary trans-complex of phospho-Parkin with cis ACT was crystallized, and structure was determined at 2.6 Å (**Extended Data Table 1**). Interestingly, in the structure of the ternary trans-complex of phospho-Parkin with cis ACT, we could observe the electron density in the ACT region (**Fig. 6C**, **Extended Data Fig. 9B**). Furthermore, Lys48, which occupied the ACT region in the ternary trans-complex of phospho-Parkin structure, was disordered in the ternary trans-complex of phospho-Parkin with cis ACT structure, similar to what is seen in the phospho-Parkin structure (**Fig. 6A**, **C**, **Extended Data Fig. 9B**).

To validate crystal structures, we compared the activity of R0RBR (141-465) and Δ Ubl-Parkin (77-465) with/without pUb, which showed that the presence of the linker containing ACT in Δ Ubl-Parkin (77-465) makes it more active compared to R0RBR (141-465) in both ubiquitination/auto-ubiquitination and Miro1 ubiquitination assays (**Extended Data Fig. 9C**). We then compared the activation of R0RBR and Δ Ubl-Parkin using pUbl (1-76) in trans. We observed that pUbl (1-76) efficiently activated Δ Ubl-Parkin (77-465); however, R0RBR (141-465) activation by pUbl (1-76) was

very poor (**Fig. 6D**, **Extended Data Fig. 9D**). Further, we tested whether pUbl-linker (1-140) with or without ACT would affect the activation of Δ Ubl-Parkin (77-465) in trans. Interestingly, ubiquitination assays performed using increasing concentrations of pUbl (1-76) or pUbl-linker (1-140), or pUbl-linker- Δ ACT (1-140, Δ 101-109) showed that Δ Ubl-Parkin activation is not affected by the linker (77-140) or ACT region in trans (**Fig. 6E**). However, compared to pUbl (1-76), pUbl-linker (1-140) showed better activation of R0RBR (141-465) (**Fig. 6F**, **Extended Data Fig. 9E**). Also, in contrast to pUbl-linker (1-140), pUbl-linker- Δ ACT (1-140, Δ 101-109) showed poor activation of R0RBR (141-465) similar to pUbl (1-76) (**Fig. 6F**, **Extended Data Fig. 9E**). However, the activity of R0RBR (141-465) complemented with pUbl-linker (1-140) was less than the activity of Δ Ubl-Parkin (77-465) complemented with pUbl (1-76) (**Fig. 6F**, **Extended Data Fig. 9E**). Overall, our data suggest that ACT can be complemented in trans; however, it is more efficient in cis.

The crystal structure of pUbl-linker (1-140) depleted Parkin (141-465, Arg163Asp, Lys211Asn) in complex with phospho-ubiquitin reveals a new ubiquitin-binding site on Parkin

In the last few years, several structures of Parkin or Parkin complexes were solved in various conditions and from different species. However, the linker (408-415) between REP element and RING2 is mostly disordered, except in structures (PDB 4I1H, 5CAW, 4ZYN) where the above region is modelled in different conformations (**Extended Data Fig. 10A**), highlighting its flexible nature. A pathogenic mutation Thr415Asn is also found in the linker (408-415), which abolishes Parkin activity. However, the role of this small linker region on Parkin remains elusive. Therefore, we decided to closely inspect all the structures solved in the present study. We noticed that in pUbl-linker (1-140) depleted Parkin (141-465, Arg163Asp, Lys211Asn) in complex with pUb structure, out of two molecules of Parkin in the asymmetric unit, one molecule of Parkin shows good electron density in the linker (408-415) region of Parkin (**Fig. 7A, B**). We further noticed that in the structure of the pUbl-linker (1-140) depleted Parkin (141-465, Arg163Asp, Lys211Asn) in complex with pUb, the linker (408-415) undergoes conformational changes compared to the previously solved apo R0RBR structure (PDB 4I1H) (**Extended Data Fig. 10B**). Thr410, Ile411 and Lys412 facing outwards in the apo R0RBR structure, are present in the core of pUbl-linker (1-140) depleted Parkin (141-465, Arg163Asp, Lys211Asn) in complex with pUb structure (**Fig. 7B**, **Extended Data Fig. 10B**). Interestingly, we noticed interactions between the linker (408-415) of Parkin and phospho-ubiquitin from the neighboring molecule in the asymmetric unit (**Fig. 7C**). The core of interactions between the Parkin linker and ubiquitin is mediated by Ile411, which is involved in hydrophobic interactions with the hydrophobic pocket of ubiquitin (**Fig. 7C**). Other interactions between Parkin and ubiquitin include ionic interactions mediated by Lys412, and His422 (**Fig. 7C**). Water-mediated interactions between linker (408-415) and ubiquitin are also seen wherein Thr410 interacts with the carbonyl group of Arg72 of ubiquitin, and Thr415 interacts with the carbonyl of Gly35 of ubiquitin (**Fig. 7C**). Furthermore, Glu409 forms a salt bridge with Lys413 (**Fig. 7C**), which could be required for maintaining the structure of the linker region for ubiquitin binding. Also, residues in the linker interacting with ubiquitin are highly conserved in Parkin across different species (**Fig. 7D**), suggesting their functional importance. Our data in **Fig. 1** suggests that RING2 remains flexible in the open state after pUbl binding in the basic patch. As the current structure is captured in the closed state, we wondered whether the linker connecting REP and RING2 may adopt an alternate conformation. The crystallization of the open state of phospho-Parkin remains challenging due to the flexible/multiple possible conformations of the REP-RING2 region. Therefore, we used AlphaFold³⁴ to predict the model of the linker region of Parkin. Interestingly, the AlphaFold model shows helical structure in the linker region of Parkin (**Extended Data Fig. 10C**), further confirming the flexible nature of this region.

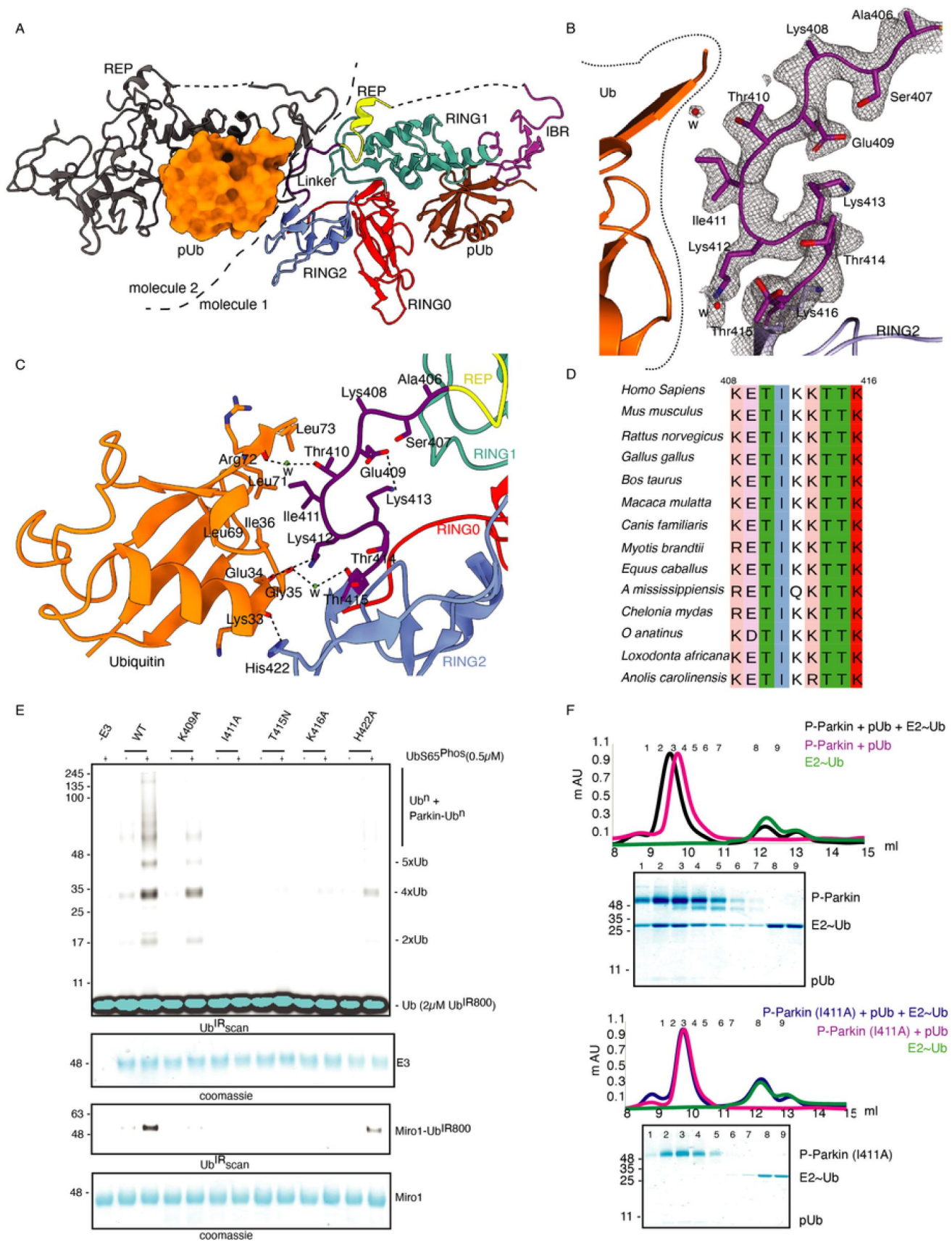


Figure 7

Linker (408-415) of Parkin binds with donor ubiquitin (Ub_{don}) of E2-Ub.

- A. The asymmetric unit of the crystal structure of pUb-linker (1-140) depleted Parkin (141-465, Arg163Asp, Lys211Asn) in complex with pUb. Domains (different colors) of Parkin molecule-1 and pUb (brown) are shown. Parkin molecule-2 (grey) and pUb (orange) are shown. The interface of two Parkin molecules is highlighted (dashed line).
- B. The 2Fo-Fc map (grey) of the linker region between REP and RING2. 2Fo-Fc map is contoured at 1.5 σ . Water molecules are represented as w.
- C. Crystal structure showing interactions between the linker (408-415) and ubiquitin. Different regions are colored as in panel A. Hydrogen bonds are indicated as dashed lines.
- D. Sequence alignment of Parkin from various species highlighting conservation in the linker (408-415) region. Residue numbers shown on top of sequence alignment are according to human Parkin.
- E. Ubiquitination assay of Parkin mutants in the linker region. The middle panel shows a Coomassie-stained loading control. The lower panel shows Miro1 ubiquitination for the respective proteins shown in the upper lane. Coomassie-stained gel showing Miro1 is used as the loading control of substrate ubiquitination assay.
- F. Size-exclusion chromatography (SEC) assay to compare the binding of E2-Ub with phospho-Parkin (upper panel) or phospho-Parkin Ile411Ala (lower panel). Assays were done using Parkin in complex with pUb. A colored key for each trace is provided. Coomassie-stained gels of indicated peaks are shown.

To validate the observations from structural analysis, we mutated these residues and compared their ubiquitination activity. In contrast to WT-Parkin, mutation of Lys409 and His422 drastically reduced Parkin activity, whilst Ile411Ala, Thr415Asn, and Lys416Ala resulted in the complete abolishment of Parkin activity (**Fig. 7E**). Further inspection revealed that although the linker region of Parkin is not conserved across different members of RBR family E3-ligases (**Extended Data Fig. 10D**), hydrophobic nature at the corresponding position of Ile411 on Parkin is conserved among various RBRs except RNF216 (**Extended Data Fig. 10D**). The crystal structures of HOIP, HOIL, HHARI, and RNF216 solved with E2~Ub³⁵⁻³⁷ show linker region interacts with donor ubiquitin (Ub_{don}) (**Extended Data Fig. 11A**). To test whether the linker between REP and RING2 of Parkin binds with donor ubiquitin (Ub_{don}), we performed binding assays using E2~Ub. Interestingly, unlike phospho-Parkin, which forms a stable complex with E2~Ub on SEC and co-elutes with E2~Ub and phospho-ubiquitin (**Fig. 7F**), Ile411Ala mutation completely abolished Parkin interaction with E2~Ub (**Fig. 7F**). Furthermore, phospho-Parkin Ile411Ala was unable to be charged by Ub-VS (**Extended Data Fig. 11B**). We also tested Parkin activity using ubiquitin mutants (Leu71Ala or Leu73Ala) which would perturb the interactions of ubiquitin and Parkin linker as suggested by structure in **Fig. 7C**. Compared to native ubiquitin, ubiquitin mutants show loss of Parkin activity (**Extended Data Fig. 11C**) which nicely corroborates with our data. Overall, our data show that the linker region between REP and RING2 interacts with donor ubiquitin and plays a crucial role in Parkin function.

Discussion

Autoinhibition of Parkin is mediated by several mechanisms. Ubl domain and REP element together block the E2 binding site on RING1^{8,22,26,27}, whereas the RING0 domain occludes the catalytic Cys431 on RING2. A few years after the discovery of Parkin autoinhibition,

various groups discovered PINK1-mediated phosphorylation of Ser65 on ubiquitin and Ubl domain of Parkin, leading to the activation of Parkin ^{10,11,12,13}. In the last few years, several structural studies have aimed to understand the conformational changes in Parkin driven by phosphorylation events that lead to Parkin activation. The structure of RING2 truncated phospho-Parkin (1-382) in complex with pUb showed that the pUbl domain of Parkin binds in the basic patch (Lys161, Arg163, Lys211) on RING0, leading to the displacement of RING2 and REP during Parkin activation ^{24,29}. Previous studies using various biophysical methods reported a K_d of ~2 μ M between Ubl with R0RBR/ Δ Ubl-Parkin; however, pUbl showed no interaction, which led to the proposed mechanism suggesting phosphorylation of Ubl resulting in the displacement of the Ubl domain leading to activation of Parkin ^{26,27}.

Our data show that RING2 and pUbl compete for binding on the basic patch of RING0 (**Fig. 1**). Our data also shows that RING2 and REP displacement after Parkin phosphorylation is transient; RING2 and REP return to their original position after the removal of the pUbl (**Fig. 1**). Our data explains that due to the net high concentration of the fused domain (RING2 or pUbl), and competitive mode of interaction, binding/displacement of pUbl/RING2 domain in trans couldn't be observed in the previous studies. However, untethering pUbl/RING2 overcomes the latter issue, and trans interaction between Parkin molecules can be observed. By untethering the linker between RING2 and IBR, after pUbl binding, the displaced RING2 is no longer able to return to the RING0 pocket, thus stabilizing the binding of pUbl on the basic patch of RING0 (**Fig. 1**). Untethered RING2 leads to a strong affinity between phospho-Ubl and core of Parkin with K_d around 1 μ M (**Fig. 2**), which is also supported by complex formation on SEC/SEC-MALS using phospho-Parkin and Parkin (**Fig. 2,3**). The dynamic nature of pUbl binding and RING2 displacement might play a crucial role in the cellular context, where the binding of pUbl with yet unknown auxiliary proteins or dephosphorylation of pUbl mediated by phosphatases may negatively regulate the Parkin activity (**Fig. 8**) and needs to be further explored.

A feedforward control mechanism is suggested in the PINK1-Parkin pathway wherein PINK1-dependent phosphorylation of ubiquitin and Parkin leads to Parkin activation on mitochondria ^{11,15,16,38,39}. However, biophysical studies aimed to understand Parkin activity did not show any dimerization of Parkin or Parkin-Parkin association in trans ^{19,30}. Our data demonstrate phospho-Parkin and WT-Parkin can form a stable complex to establish the Parkin dimerization in trans (**Fig. 3**). We further show that phospho-Parkin can activate WT-Parkin in trans, suggesting the major mode of Parkin autoinhibition is mediated by RING0 blocking the RING2 domain. Our data suggest an additional feedforward activation model of Parkin wherein fully-activated Parkin (phospho-Parkin bound with pUb) molecules can activate partially-activated Parkin (WT-Parkin bound with pUb) molecules mediated by interactions between pUbl and RING0 in trans (**Fig. 8**). The latter can be useful in the context of healthy carriers of heterozygous mutations on Parkin. The critical role of pUbl supports data showing the importance of Ubl phosphorylation in vivo as demonstrated by the discovery of Parkinson's patients associated with homozygous Ser65Asn mutations ⁴⁰. This data also highlights the importance of various Parkin isoforms that have been identified (**Extended Data Fig. 12**), especially the ones that lack Ubl domain or REP-RING2 domains, as they can complement each other using our proposed trans model in **Fig. 8**.

ACT is proposed to have a role in Parkin activation, as it was shown that the deletion/mutation of ACT leads to the loss of Parkin activity ²⁴. We demonstrate that ACT plays a key role due to its inherent capacity to bind with the RING0 pocket. Unlike other functional mutations on Parkin affecting interaction with E2 or Ub_{don}, ACT deletion does not affect binding with E2~Ub. We show that ACT plays a crucial role in enzyme kinetics and only slows the Parkin activity, possibly due to affecting the inherently dynamic nature of RING2 (**Fig. 5**). Furthermore, we also demonstrate that ACT can be complemented in trans; however, ACT on a cis molecule is more effective (**Fig. 6**). Also, Lys48 of the Ubl domain of Parkin, previously suggested to release autoinhibition ⁸, undergoes conformational changes between structures of trans-complex of phospho-Parkin using

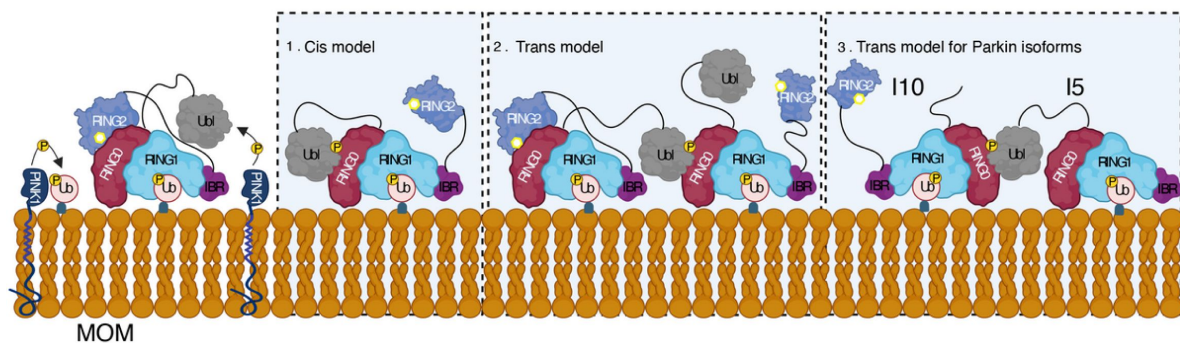


Figure 8

Model showing different modes of Parkin activation.

The cis-activation model uses binding of pUb in the same molecule, thus resulting in the displacement of RING2 (1). The trans-activation model uses the binding of pUb of fully-activated parkin (phospho-parkin complex with pUb) with partially-activated parkin (WT-parkin and pUb complex), thus resulting in the displacement of RING2 in trans (2). Recruitment and activation of parkin isoforms lacking Ub (Isoform 10) or RING2 domain (Isoform 5), thus complementing each other using trans-activation model (3). Catalytic cysteine on RING2 is highlighted.

cis or trans ACT suggesting it might play a crucial role in Parkin regulation (Fig. 6). The role of ACT demonstrated in the present study also explains why ACT is conserved only in chordates where it might be evolved to enhance the activity of Parkin.

The linker connecting IBR and RING2 of Parkin comprises two components: a REP element (391-405) and a flexible linker (408-415). Various Parkin structures solved so far show REP element blocking the E2 binding site on RING1; however, linker (408-415) remains flexible in most structures, and its role remains elusive. Interestingly, pathogenic mutation Thr415Asn has been shown to defect the E3 ligase activity of Parkin⁸. Also, using peptide array analysis, Chaugule and colleagues proposed a Parkin Ubl/ubiquitin-binding (PUB) site in the C-terminal domain of Parkin⁸. Here, we demonstrate that the linker (408-415) interacts with donor ubiquitin (Ub_{don}) of E2~Ub (Fig. 7). Although the linker between IBR-RING2 is not conserved across RBR family E3-ligases, the core of interaction between the linker and Ub_{don} is mediated by hydrophobic residue in the linker region. The linker between IBR-RING2 of Parkin is present in an extended conformation, leading to IBR and RING2 occupying diagonally opposite conformation, which is quite similar to what is seen in HOIP RBR and E2~Ub complex structure (Fig. 7, Extended Data Fig. 11A³⁵). However, the recent structures of RBR family E3-ligases (HHARI, RNF216, HOIL-1)^{36,37} show a kinked conformation of the linker connecting IBR-RING2 (Extended Data Fig. 11A). Interestingly, the kink in the linker region plays a crucial role in bringing RING2 to the catalytically feasible state in a cis molecule (Extended Data Fig. 11A). In contrast, catalytic feasibility under the extended conformation of the linker is only possible in trans (Extended Data Fig. 11A). This conformational change that the linker region adopts may also play a key role in the function of RBR family E3-ligases. The conformational flexibility in the linker (408-415) region of Parkin is also supported by the various secondary structures (disordered<->loop<->helix) (Fig. 7, Extended Data Fig. 10). The latter flexibility in the linker region could explain the dynamic nature of RING2 between the presence or absence of pUbl in Parkin. Previous data suggesting the opening of RING2 after the addition of E2~Ub in RORBR³⁰, wherein RING2 is in the closed state, suggests that the conformation of donor ubiquitin and linker captured in the present structure might be one of the possible intermediates as mentioned above. The latter observation also suggests that conformational changes might be induced in the linker region after binding with donor-ubiquitin and needs further investigation. Although the regulatory mechanism varies across RBR family E3-ligases, the catalytic core (IBR-RING2) undergoes similar conformational changes, leading to a unified catalysis mechanism in various RBR family E3-ligases.

Overall, this study answers several critical questions related to Parkin activation and regulation, which could aid in the development of small-molecule regulators against Parkin to treat PD.

Declarations

Authors' contributions

DRL performed all experiments, solved structures, and analyzed the data. SD purified mutant proteins and performed SEC assays in Fig. 6, and did the sequence alignment analysis. OA performed and analyzed IF experiments under the supervision of MM. AP; IF imaging and quantification. PS performed ITC experiments. AK conceived, designed, and supervised the research, refined structures, analyzed the data, and wrote the manuscript with input from all authors.

Conflict of interest

MM. is a member of the Scientific Advisory Board of Montara Therapeutics Inc and scientific consultant to MSD UK.

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Methods

Molecular Biology

The human *PARK2* gene optimized for bacterial expression of FL-Parkin was cloned in the pET15b vector. Various Parkin mutations used in the present study were made using site-directed mutagenesis (SDM). TEV cleavage site (ENLYFQS) was substituted in the Parkin construct between the 382nd - 388th as described in [24](#), and an HRV 3C site (LEVLFQGP) was inserted between residue 140th and 141st residues using site-directed mutagenesis. Ubl (expressing 1-76th amino acids of Parkin) and Ubl-linker (expressing 1-140th amino acids of Parkin) constructs were generated by introducing a stop codon after the 76th and 140th amino acids, respectively, in the FL-Parkin construct. Ubl-linker-ΔACT (1-140, Δ101-109) construct was made on Ubl-linker using SDM. Miro1 (expressing 181st - 592nd amino acid) was amplified from the cDNA of the HEK293T cell line using Phusion polymerase (NEB) and cloned into the pGEX-6P1 vector using EcoRI and BamHI restriction enzymes. To generate fluorescently labeled ubiquitin, ubiquitin (residues 2-76) was cloned in a pGEX-6P vector with an overhang expressing GPLCGS at the n-terminal of ubiquitin. For the generation of ubiquitin-3Br protein, the ubiquitin gene (residues 1-75) was cloned in the pTXB-1 vector. *Pediculus humanus corporis* PINK1 (115 - 575) was a gift from David Komander [43](#) (Addgene plasmid # 110750). Ube1 was a gift from Cynthia Wolberger [44](#) (Addgene plasmid # 34965).

Protein purification

Parkin constructs were expressed in *Escherichia coli* BL21(DE3)pLysS cells. Cells were grown until OD₆₀₀ reached 0.4; the temperature was reduced to 16 °C, and protein was induced by adding 50 μM IPTG, and media was supplemented with 200 μM ZnCl₂. Cells were left to grow overnight at 16 °C. Cells were harvested and lysed using sonication in lysis buffer (25 mM Tris pH 7.5, 200 mM NaCl, 5 mM Imidazole, 1 mM β-mercaptoethanol, and 100 μM AEBSF). Protein was purified over Ni-NTA resin. His-Sumo tag was removed using SENP1 protease. Protein was further purified over Hi-Trap Q HP column (GE Healthcare) followed by a gel-filtration column pre-equilibrated with storage buffer (25 mM Tris pH 7.5, 75 mM NaCl, 250 μM TCEP). Other proteins were also purified using similar protocols. PhPINK1 was purified as published before [43](#).

Isothermal Titration calorimetry

Isothermal titration calorimetry experiments were performed using Nano ITC (TA instruments). All titrations were performed at 25°C in a buffer containing 50 mM HEPES (pH 7.5), 100 mM NaCl, and 250 μ M TCEP. In **Fig. 2A**, experiments were done using 350 μ M of P-Parkin (K211N) in the syringe and 25 μ M of Δ Ubl-Parkin in the cell. In **Fig. 2B**, experiments were done using 220 μ M of Δ Ubl-Parkin (RING2 untethered with TEV) in the syringe and 40 μ M of P-Parkin (K211N) in the cell. Results were analyzed in NanoAnalyze software using an independent mode of a binding model.

Ubiquitination assays

Ubiquitination assays were performed using fluorescently labeled ubiquitin. Ubiquitin labeling was done using DylightTM 800 Maleimide (Thermo Scientific), as mentioned in [26](#), using the manufacturer's specifications. Ubiquitination reactions were performed at 25 °C for 40 minutes in 25 mM Tris pH 7.5, 50 mM NaCl, 10 mM MgCl₂, and 0.1 mM DTT, 10 mM ATP. In all reactions, 25 nM Ube1, 250 nM Ubch7 (E2), 1 μ M of E3, and 2 μ M of Ub^{IR800} were used in 20 μ l of the total reaction volume. 0.5 μ M of Ub or pUb was used as an allosteric activator for the experiments in **Extended Data Fig. 4B**, **Fig. 7E**, and **Extended Data Fig. 9C**. Increasing concentrations of pParkin (T270R, C431A) (1 μ M, 2 μ M, 4 μ M, and 8 μ M) were used as trans activators in **Fig. 3D**. The transactivation experiments using pUbl, pUbl-linker, and pUbl-linker- Δ ACT were carried out in increasing concentrations of 4 μ M, 8 μ M, and 16 μ M in **Fig. 6D, E, F**. Miro1 substrate ubiquitination reaction was done at 25 °C for 20 minutes with 5 μ M Miro1 and 0.5 μ M of E3. Other conditions were the same as mentioned above for ubiquitination/autoubiquitination assay. The reactions were quenched by SDS loading dye and heated at 95 °C for 5 mins. The samples were resolved on gradient SDS-PAGE and analyzed using Li-COR[®] Odyssey Infrared Imaging System. Each assay was repeated at least three times. ImageJ software was used to quantification of ubiquitination. Bar plots and statistical analysis were done using R.

Cell culture transfection and microscopy experiment

HeLa cells were cultured in Dulbecco's modified Eagle's medium (DMEM; Gibco) containing 10% (vol/vol) FBS, 1% Pen/Strep and 1% L-Glutamine at 37°C under an atmosphere of 5% CO₂. 24-well cell culture plate (35000 cells/well) were used to seed cells onto borosilicate cover glasses (VWR 631-0148). The following plasmids were generated by MRC Reagent & Services and used to assess Parkin translocation: GFP-Parkin (DU23318), mCherry-Parkin (DU77708), mCherry-Parkin-S65A (DU77709), GFP-Parkin-C431F (DU77645), GFP-Parkin-K211N-C431F (DU77659) and GFP-Parkin-H302A-C431F (DU77713). Transfections were carried out the day after seeding and plasmids were mixed with PEI (PEI MAX-Polyscience, 24765-1) at 1:5 ratio in Opti-MEM (Gibco). DNA/PEI mix was left 45 minutes at room temperature, then added to the cell cultures and incubated for 48h before CCCP treatment (10 μ M for 1h). For immunostaining, cells were fixed with 4% (wt/vol) paraformaldehyde in PBS for 20 min at room temperature and permeabilized with blocking buffer, containing 3% (wt/vol) Donkey serum and 0.2% (vol/vol) Triton X-100 in PBS for 1 h. Cells were incubated with TOMM20 (ab186735) primary antibody overnight at 4 °C, followed by incubation with the anti-mouse Alexa Fluor 405 secondary antibody (ThermoFisher, A-48258) for 1h at room temperature. After 3 washes in PBS, and a rinse in milliQ water, the cover glasses were mounted onto slides using Vectashield mounting medium (Vector Laboratories, H-1000). Microscopy was performed on an LSM 880 laser scanning confocal microscope (ZEISS; Plan-Apochromat 63x/NA 1.4) using ZEISS Zen Software. Colocalization was assessed using Volocity Software (version 6.3, Quorum Technologies) and determined as Pearson's correlation coefficient for mitochondrial colocalization of GFP and the mitochondrial marker TOMM20. Images were processed using ImageJ software version 1.51 (100).

Purification of phospho-Ubiquitin (pUb)-3Br

pUb-3Br was purified as published before ^{28,45}. Briefly, Ubiquitin (1-75)-Mxe-intein-chitin binding domain was expressed using *Escherichia coli* BL21(DE3) cells using a pTXB-1 vector. Cells were induced at 0.8 O.D. with 250 μ M IPTG and incubated at 22 °C for 12 hrs. Cells were lysed in lysis buffer (20 mM Na₂HPO₄ pH 7.2, 200 mM NaCl, 0.1 mM EDTA) and purified using Chitin resin (NEB). The resin was incubated with cleavage buffer (20 mM Na₂HPO₄ pH 6.0, 200 mM NaCl, 50 mM MESNa, 0.1 mM EDTA) overnight to elute the protein. The eluted protein was reacted with 3-Bromopropylamine (Sigma) at 25 °C for 4 hrs. The reacted protein was purified over Hiload 16/600 Superdex 75pg column (GE Healthcare) pre-equilibrated with 1X PBS. The fractions containing Ub-3Br were concentrated and phosphorylated using PhPINK1. pUb-3Br was purified over Hiload 16/600 Superdex 75pg column pre-equilibrated with Parkin storage buffer.

Synthesis and purification of UbCH7~Ub

The reaction containing 500 μ M of UbCH7 (Cys17Ser/Cys86Ser/Cys137Ser), 15 μ M of Ube1, and 2.5 mM of 6xHis-Ub in charging buffer (50 mM HEPES pH 7.5, 150 mM NaCl, 10 mM MgCl₂, 10 mM ATP) was incubated at 37 °C for 18 hrs. The progress of the reaction was monitored over SDS-PAGE. The reacted sample was passed through Ni-NTA resin to capture His-Ub and UbCH7~Ub (His), and the eluted fraction was purified over Hiload 16/600 Superdex 75 pg column (GE Healthcare). Fractions containing UbCH7~Ub were pooled together and stored for further use.

Preparation of Parkin complexes for crystallization

In the present study, Parkin complexes with pUb were captured using pUb-3Br. To capture Parkin complexes with pUb-3Br, human Parkin constructs were mutated to include Gln347Cys, as published before ²⁸, in various constructs for crystallization experiments. For crystallization of pUbl-linker (1-140) depleted Parkin (141-465) and pUbl-linker (1-140) depleted Parkin (141-465, Arg163Asp, Lys211Asn) recombinant constructs containing FL-Parkin (HRV 3C site between 140th – 141st residue, TEV site between 382nd – 383rd, Gln347Cys) and FL-Parkin (HRV 3C site between 140th – 141st residue, Arg163Asp, Lys211Asn, Gln347Cys) respectively, were made. Proteins were expressed and purified as above. Purified proteins were mixed with pUb-3Br, and Parkin was phosphorylated using PhPINK1 in a phosphorylation buffer containing 5 mM ATP along with pUb-3Br. GST-HRV 3C protease was added (1:50), and proteins were left overnight at 4 °C. The proteins were passed through affinity chromatography to remove GST-HRV 3C protease and PhPINK1. Flow-through was further purified over a gel-filtration column. Fractions containing R0RBR with pUb were pooled together and used for crystallization.

Ternary trans-complex of phospho-Parkin (1-140 + 141-382 + pUb) was made using Parkin (Lys211Asn, HRV 3C site between 140th – 141st residue) construct as the donor of pUbl-linker, and R0RBR (141-465, TEV site between 382nd – 383rd, Gln347Cys) construct as the acceptor of pUbl-linker. Purified Parkin (Lys211Asn, HRV 3C site between 140th – 141st residue) was phosphorylated using PhPINK1 as above. Purified R0RBR (141-465, TEV site between 382nd – 383rd, Gln347Cys) was treated with His-TEV, and His-TEV was removed over Ni-NTA resin. 2-fold molar excess of phospho-Parkin (Lys211Asn, HRV 3C site between 140th – 141st residue) was mixed with R0RBR (Gln347Cys, untethered RING2). The complex containing phospho-Parkin (Lys211Asn, HRV 3C site between 140th – 141st residue) and R0RB (141-382, Gln347Cys) was purified over Hiload 16/600 Superdex 200pg column pre-equilibrated with Parkin storage buffer. The latter complex was mixed with pUb-3Br and treated with 3C protease. Protein was further purified over Hiload 16/600 Superdex 75pg column pre-equilibrated with Parkin storage buffer. Fractions containing ternary trans-complex of phospho-Parkin (1-140 + 141-382 + pUb) were pooled together, concentrated, and used for crystallization.

Ternary trans-complex of phospho-Parkin with cis ACT (1-76 + 77-382 + pUb) was made using the Ubl domain of Parkin (1-76) and Δ Ubl-Parkin (77-465, TEV site between 382nd – 383rd, Gln347Cys). Δ Ubl-Parkin (77-465, TEV site between 382nd – 383rd, Gln347Cys) was treated with His-TEV, and His-TEV was removed over Ni-resin. A 3-fold molar excess of the pUbl domain was mixed with Δ Ubl-Parkin (Gln347Cys, untethered RING2). The complex containing pUbl and Δ Ubl-Parkin (77-382, Gln347Cys) was purified over Superdex 75increase 10/300 GL column pre-equilibrated with Parkin storage buffer. The latter trans-complex of Parkin with cis ACT (1-76 + 77-382) was mixed with pUb-3Br and purified again over Superdex 75increase 10/300 GL column pre-equilibrated with Parkin storage buffer. Fractions containing ternary trans-complex of phospho-Parkin with cis ACT (1-76 + 77-382 + pUb) were pooled together, concentrated, and used for crystallization.

RORBR (141-465, Gln347Cys, TEV site between 382nd – 383rd, Gln347Cys) was purified as stated above. After treatment with TEV, TEV was depleted using Ni-NTA resin, and untethered RORBR was purified using Hiload 16/600 Superdex 75pg column pre-equilibrated with Parkin storage buffer.

Crystallization and structure determination

Initial crystals of pUbl-linker (1-140) depleted Parkin (141-465) complex with pUb-3Br appeared in 1.6 M Ammonium sulfate, 0.1 M MES monohydrate pH 6.5, and 10% v/v 1,4-Dioxane of HR112 screen (Hampton Research) at 4°C. Seeding was done to grow bigger crystals in the same condition. The mother liquor containing 20% (v/v) of glycerol was used as a cryoprotectant for freezing crystals in liquid nitrogen. pUbl-linker (1-140) depleted Parkin (141-465, Arg163Asp, Lys211Asn) complex with pUb-3Br crystals appeared in 0.15 M Potassium bromide, and 30% w/v Polyethylene glycol monomethyl ether 2,000 of Index screen (Hampton research) at 18°C. The mother liquor containing 20% (v/v) of PEG 400 was used as a cryoprotectant for freezing crystals in liquid nitrogen. Crystals of ternary trans-complexes of phospho-Parkin were obtained in 0.3 M Sodium nitrate, 0.3 Sodium phosphate dibasic, 0.3 M Ammonium sulfate, 0.1 M Tris (base) & BICINE (pH 8.5), 25% v/v MPD, 25% w/v PEG 1000, and 25% w/v PEG 3350 of Morpheus screen (Molecular dimensions). Good quality crystals were grown at 18 °C using microseeding. The mother liquor containing 10% (v/v) of glycerol was used as a cryoprotectant for freezing crystals in liquid nitrogen. Crystals of untethered RORBR c were grown in 0.1 M HEPES, pH 7.5, 8% PEG 4000, 10% isopropanol, and 0.1 M BaCl₂ at 4 °C. The mother liquor containing 20% (v/v) glycerol was used for vitrification.

Data were collected at the European Synchrotron Radiation Facility (ESRF), Grenoble, France. Data were processed using XDS [46](#). Scaling was done using Aimless, and the structure was determined by molecular replacement using Phaser as implemented in CCP-7.1 [47](#). Structures of pUbl-linker (1-140) depleted Parkin (141-465) or pUbl-linker (1-140) depleted Parkin (141-465, Arg163Asp, Lys211Asn) in complex with pUb-3Br were solved by using Pediculus Parkin structure in complex with phospho-ubiquitin (PDB 5CAW) as a search model. Structures of ternary trans-complex of phospho-Parkin were solved using phospho-Parkin structure (PDB 6GLC) as a search model. Untethered RORBR structure was determined using RORBR structure (PDB 4I1H) as a search model. The initial model was built and refined using coot [48](#) and refmac5 [49](#).

Purification of phosphorylated proteins

PhPINK1 was used to phosphorylate various Parkin variants used in the study. Phosphorylation buffer contains 50 mM Tris pH 8.5, 100 mM NaCl, 10 mM MgCl₂, 10 mM DTT, and 10 mM ATP. The reactions were performed at 25 °C for 4 hrs. Phosphorylation status was checked using Phos-Tag (FUJIFILM) analysis as per the manufacturer's protocol. PINK1 was depleted by affinity chromatography upon completion of the reaction. The phosphorylated proteins were further purified over a gel-filtration column.

Parkin phosphorylation assay

Parkin phosphorylation assay was performed using 5 μ M Parkin and 0.25 μ M PINK1 in phosphorylation buffer at 25 °C for 15 mins. Increasing concentrations (20 μ M, 40 μ M, and 80 μ M) of pUbl or pUb were added with Parkin to check their effect on Parkin phosphorylation. The samples were analyzed using Phos-Tag (FUJIFILM) analysis using the manufacturer's protocol.

Size-exclusion chromatography

For RING2 or Ubl displacement/binding assays, HRV-3C cleavable and TEV cleavable constructs of Parkin were purified and phosphorylated as above. TEV and HRV 3C were added at the molar ratio (protease: Parkin) of 1:5 and 1:15, respectively. After incubation with respective proteases, proteins were purified using affinity chromatography to remove proteases from Parkin. The proteins were loaded onto Superdex 75increase 10/300 GL column, and fractions were analyzed using SDS-PAGE.

For the trans-complex assays, phospho-Parkin variants were added in 2-fold molar excess. Also, in all trans-complex assays, the TEV site between IBR and RING2 was present only on the target molecules. Furthermore, before complex formation, TEV was removed by affinity chromatography. Proteins were incubated for 30 minutes at 4 °C before loading onto Superdex 75increase 10/300 GL column. Fractions were analyzed using SDS-PAGE.

For SEC assay to analyze Parkin interaction with E2~Ub, 10 μ M of phospho-Parkin/phospho-Parkin- Δ ACT/phospho-Parkin Ile411Ala was pre-incubated with 15 μ M of pUb, followed by the addition of 20 μ M of E2~Ub. Proteins were incubated for 1 hr at 4°C before injecting onto Superdex 75increase 10/300 GL column. Fractions were analyzed using SDS-PAGE to check the complex formation.

Sec mals

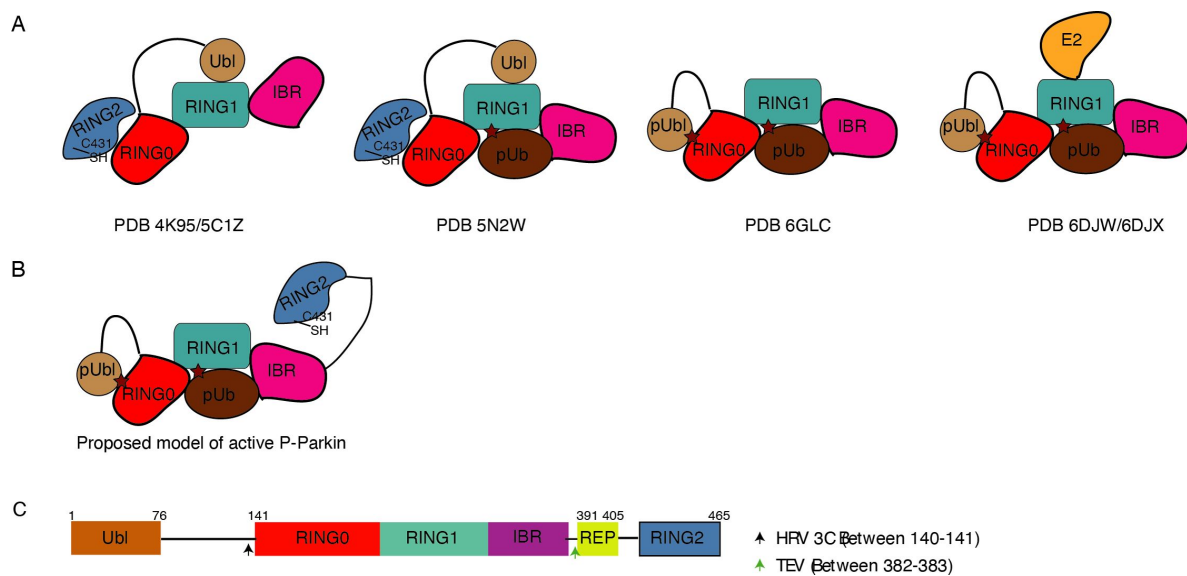
Size-exclusion chromatography was performed with inline multi-angle light scattering using the Viscotek SEC-MALS 20 system. Protein at 4-6 mg/mL (100 μ L) was loaded on P2500-P4000 columns (Malvern) at a flow rate of 0.3 mL/min in buffer containing 20 mM Tris-HCl pH 7.5, 75 mM NaCl, 0.25 mM TCEP. The data were analyzed using OmniSEC 5.11 software.

Supplementary Files

	ternary trans-complex of phospho-parkin with pUb	pUbl-linker depleted parkin (141-465) complex with pUb	Untethered R0RBR	ternary trans-complex of phospho-parkin with cis ACT and pUb	pUbl-linker depleted parkin (141-465, R163D, K211N) complex with pUb
Data collection					
Resolution range	34.30 - 1.92 (1.98 - 1.92)	39.15 - 3.3 (3.41 - 3.3)	48.28 - 2.9 (3.004 - 2.9)	35.84 - 2.6 (2.69 - 2.6)	37.47 - 2.35 (2.43 - 2.35)
Space group	P 32 2 1	P 64 2 2	C 2 2 21	P 32 2 1	P 1 21 1
Cell dimensions					
<i>a</i> , <i>b</i> , <i>c</i> (Å)	83.804, 83.804, 105.033	187.805, 187.805, 141.857	86.672, 132.579, 64.692	82.764, 82.764, 103.494	45.45, 76.426, 114.329
α , β , γ (°)	90, 90, 120	90, 90, 120	90, 90, 90	90, 90, 120	90, 100.485, 90
Total reflections	186235 (17323)	373278 (38759)	32244 (3362)	170918 (17328)	97886 (9139)
Unique reflections	32991 (3234)	22705 (2115)	8492 (843)	13023 (1273)	31073 (3105)
Multiplicity	5.6 (5.4)	16.4 (17.4)	3.8 (4)	13.1 (13.6)	3.2 (2.9)
Completeness (%)	99.55 (99.35)	91.65 (92.19)	98.84 (99.29)	99.17 (98.82)	96.28 (96.61)
<i>I</i> / σ (<i>I</i>)	14.09 (1.41)	11.75 (0.72)	11.83 (3.78)	17.69 (1.94)	12.45 (2.55)
Wilson B-factor	43.26	132.82	49.22	46.5	40.73
R-merge	0.05746 (0.9399)	0.1928 (3.632)	0.09615 (0.3465)	0.1416 (1.696)	0.0735 (0.52)
CC1/2	0.993 (0.738)	0.998 (0.48)	0.993 (0.921)	0.999 (0.763)	0.996 (0.695)
Refinement					
Reflections used in refinement	32964 (3229)	20836 (2053)	8492 (843)	12933 (1258)	31062 (3105)
R-work/R-free	0.2031/0.2368	0.2360/0.2750	0.2180/0.2438	0.2141/0.2355	0.1952/0.2119
No of Atoms					
macromolecules	2937	5457	2396	3006	6033
ligands	19	59	21	30	37
solvent	96	2	19	52	258
RMS deviations					
Bond length (Å)	0.008	0.009	0.008	0.009	0.007
Bond angles (°)	1.20	1.24	1.64	1.20	1.17
B-factors					
macromolecules	61.01	150.78	44.37	64.69	44.87
ligands	70.26	196.7	45.47	79.76	48.68
solvent	60.53	118	38.72	65.63	47.36
Accession code	8IKM	8IK6	8JWV	8IKT	8IKV

Table 1

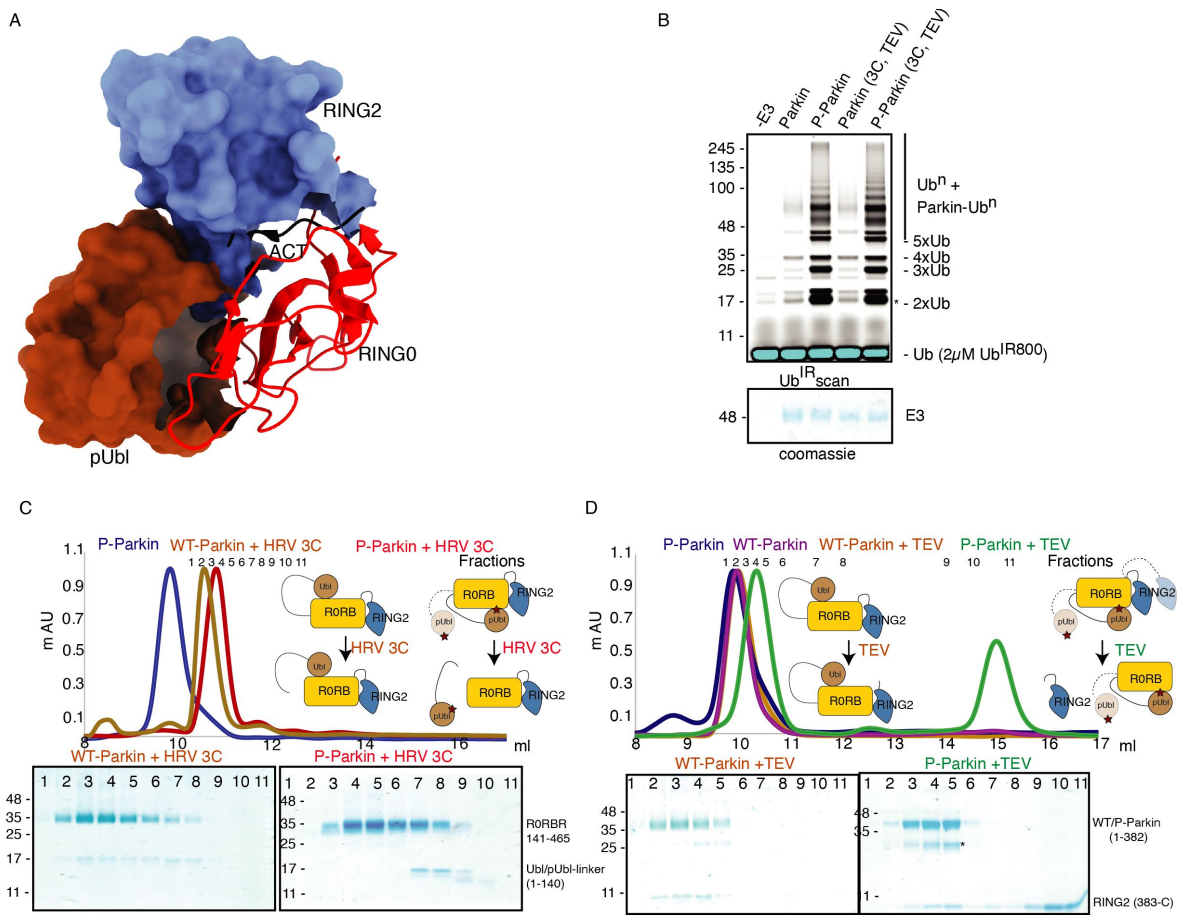
Data collection and Refinement statistics



Extended Data Figure 1

Summary of Parkin conformations/model.

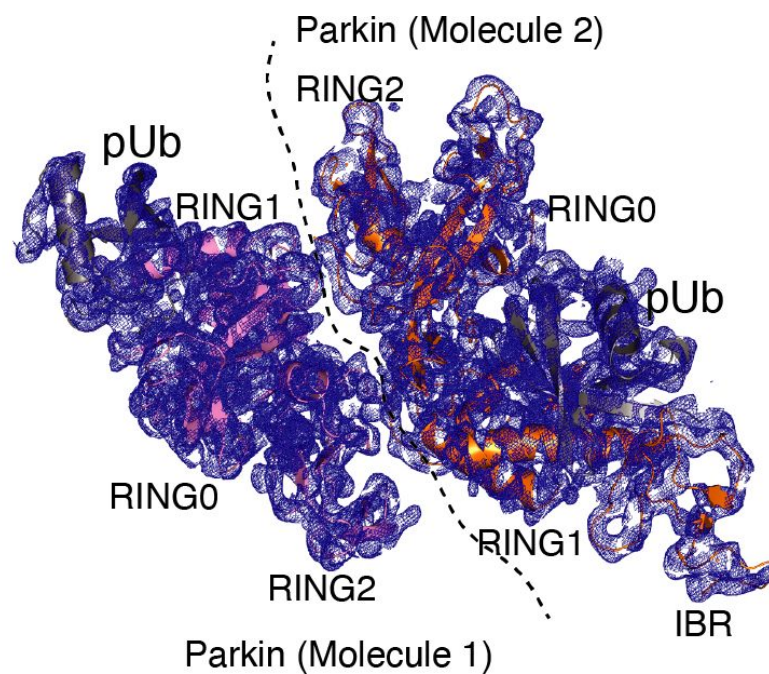
A. Conformation of Parkin observed in various crystal structures solved so far B. Proposed conformation of phospho-Parkin [24](#), [29](#) C. Schematic representation of Parkin domains. HRV 3C and TEV sites incorporated in the Parkin construct are marked with black and green arrows, respectively



Extended Data Figure 2

pUbl and RING2 have a competitive mode of binding on RING0.

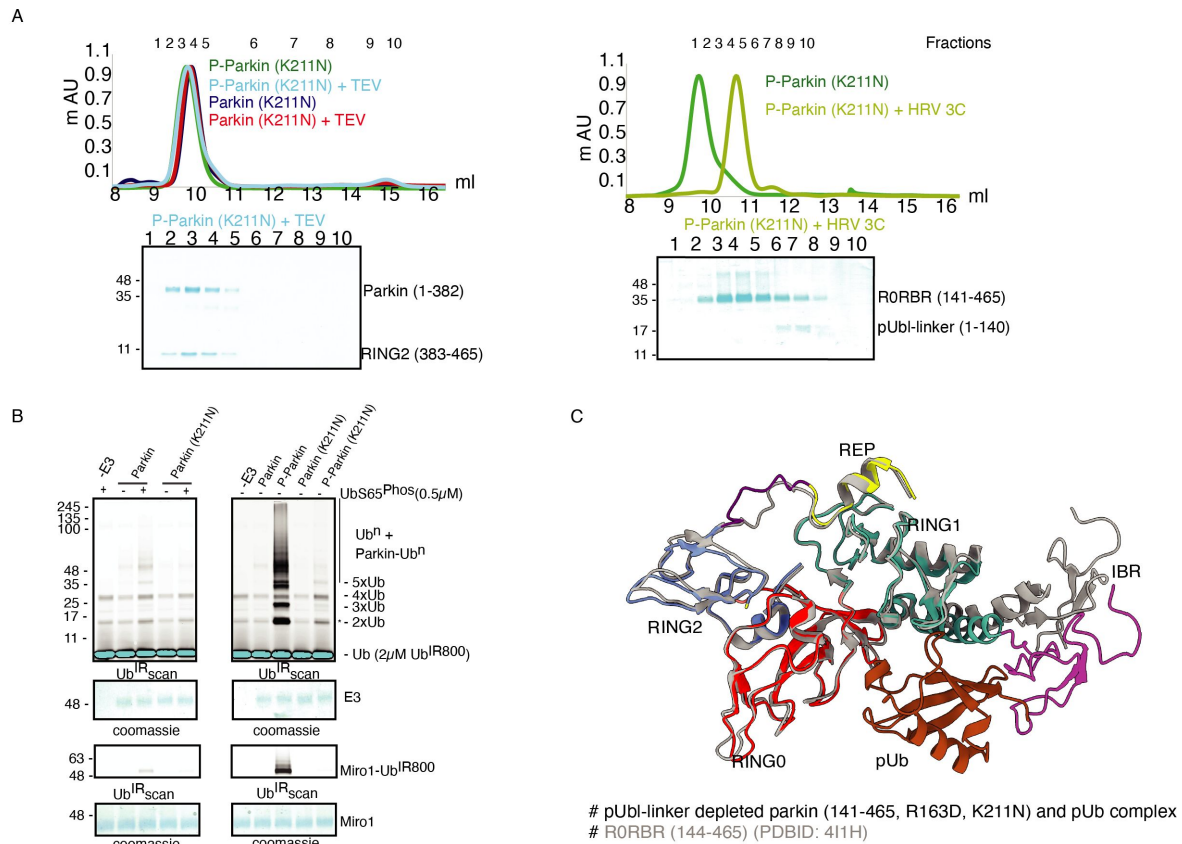
A. Superimposition of the structure of WT-Parkin (PDB 5C1Z) and phospho-Parkin (PDB 6GLC) structure. RING2 (blue), pUbl (brown), RING0 (red), and ACT (black) are shown. For clarity, other domains of Parkin are not included. B. Ubiquitination assay on Parkin construct with TEV and HRV 3C sites. A non-specific, ATP-independent band is indicated (*). The lower panel shows a Coomassie-stained loading control. C. Size-exclusion chromatography (SEC) assay showing the binding/displacement of Ubl-linker (1-140) under native or phosphorylated conditions. A colored key for each trace is provided. Coomassie-stained gels of indicated peaks are shown in the lower panel. A schematic representation is used to explain SEC data. D. Size-exclusion chromatography (SEC) assay showing binding/displacement of RING2 (383-465) under native or phosphorylated conditions. Coomassie-stained gels of indicated peaks are shown in the lower panel. TEV as contamination is indicated (*).



Extended Data Figure 3

Density map of pUbl-linker (1-140) depleted Parkin (141-465) complex with pUb structure.

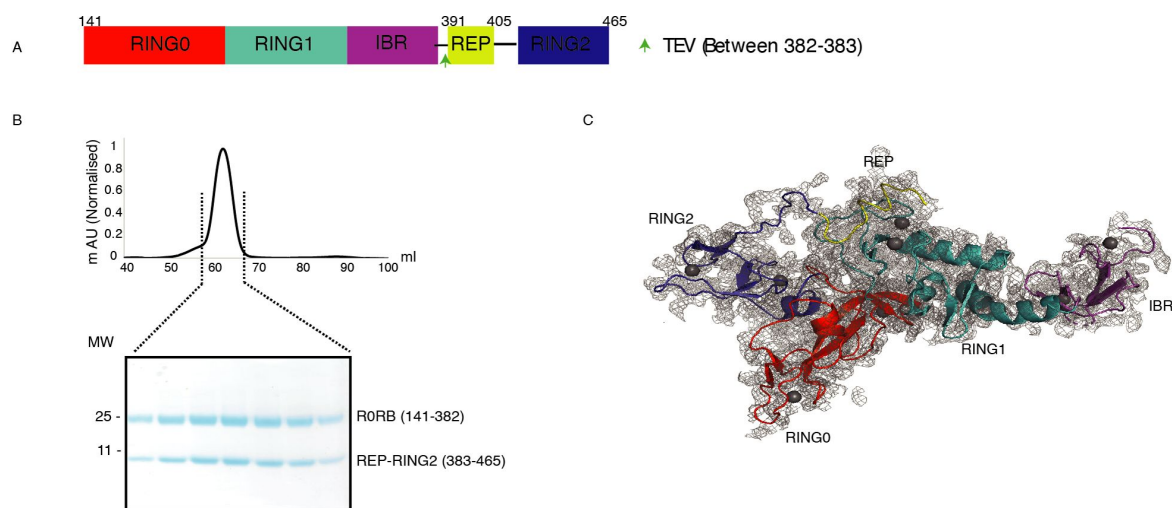
The 2Fo-Fc map (blue) of Parkin molecules (shown in different colors) in the crystal structure of the pUbl-linker (1-140) depleted Parkin (141-465) complex with pUb (grey). The 2Fo-Fc map is contoured at 1.5 σ .



Extended Data Figure 4

K211N mutation affects RING2 displacement but not pUbl.

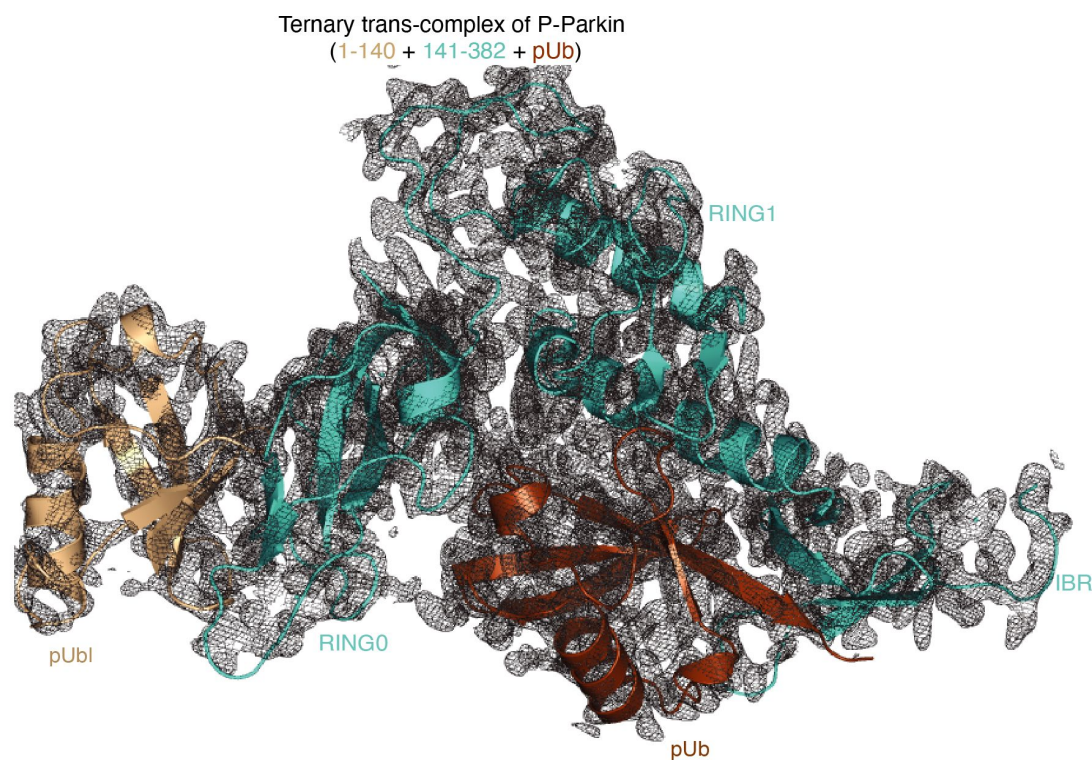
A. Size-exclusion chromatography (SEC) assay to test the displacement of RING2 (left panel) or pUbl-linker (right panel) after phosphorylation of Parkin (Lys211Asn). B. Ubiquitination assay to test the activity of Parkin Lys211Asn in the presence of pUb or using phospho-Parkin Lys211Asn. The middle panel shows a Coomassie-stained loading control. A non-specific, ATP-independent band is indicated (*). The lower panel shows Miro1 ubiquitination for the respective proteins shown in the upper lane. Coomassie-stained gel showing Miro1 is used as the loading control of substrate ubiquitination assay. C. Crystal structure of pUbl-linker (1-140) depleted Parkin (141-465, Lys163Asp, Lys211Asn) in complex with pUb (brown), different domains of Parkin are colored as panel E, superimposed apo R0RBR structure (PDB 4I1H) is shown in grey.



Extended Data Figure 5

Parkin treatment with TEV does not affect native interactions.

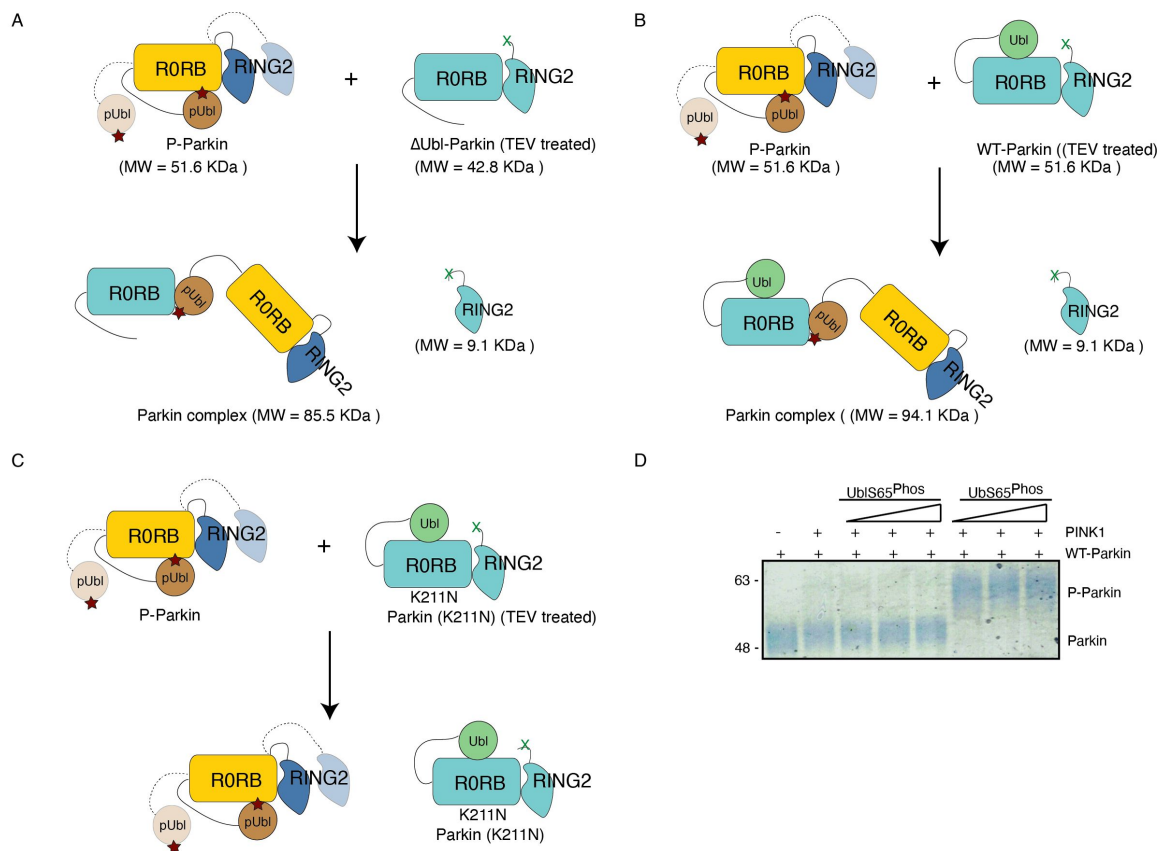
A. Schematic representation of the R0RBR construct. TEV sites incorporated in the Parkin construct are marked green arrows, respectively B. Purification of untethered R0RBR Parkin over Hiload 16/600 Superdex 75pg column. Fractions from the highlighted region of the peak were loaded on SDS-PAGE (lower panel). C. Crystal structure of untethered R0RBR Parkin. The 2Fo-Fc map is shown in grey. Parkin domains are shown in different colors.



Extended Data Figure 6

Electron density map of the ternary trans-complex of Parkin.

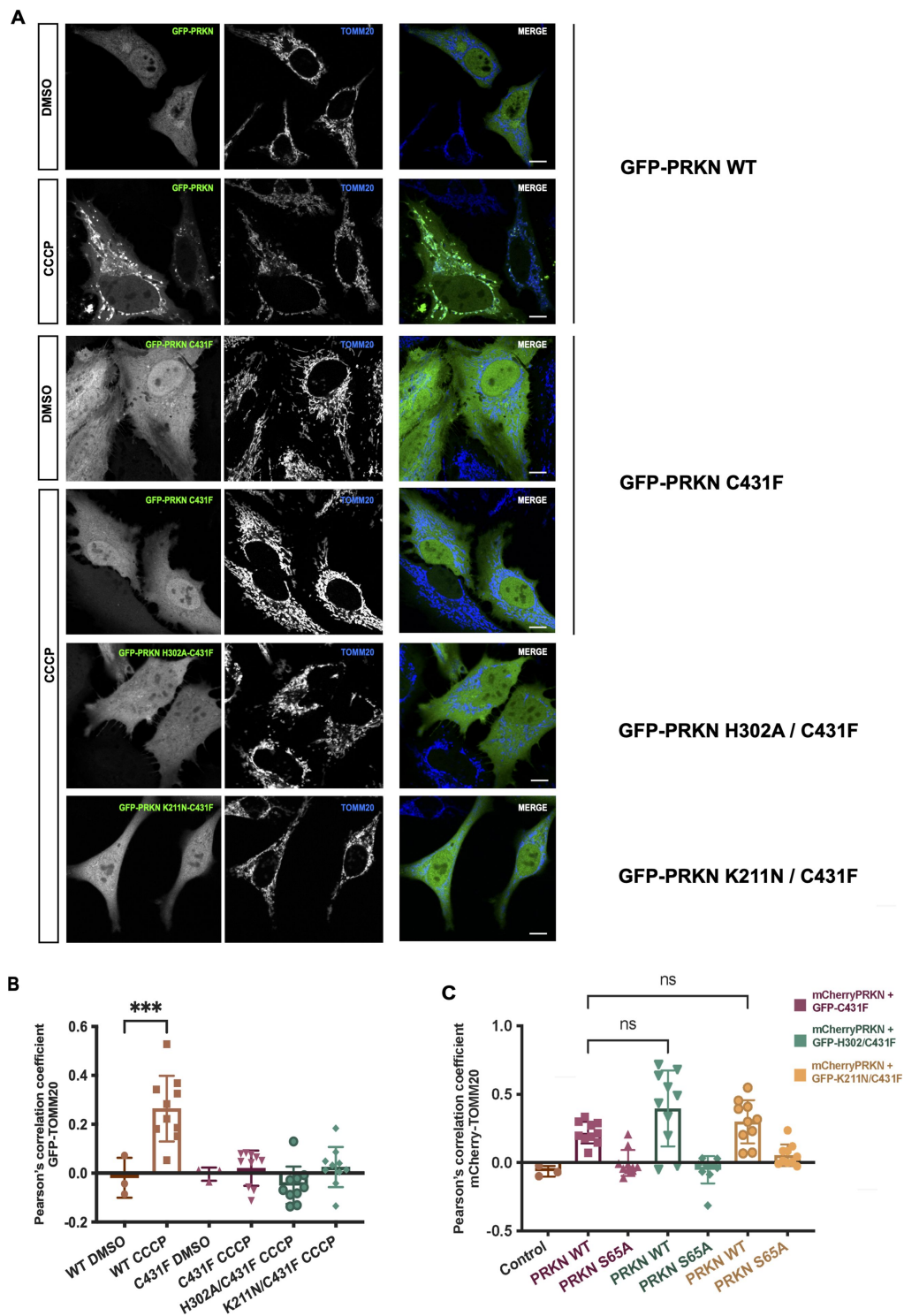
The 2Fo-Fc map (grey) of the ternary trans-complex of phospho-Parkin (1-140 (wheat) + 141-382 (cyan) + pUb (brown)). The 2Fo-Fc map is contoured at 1.5 σ .



Extended Data Figure 7

Phosphorylation of Parkin leads to the association of Parkin molecules in trans.

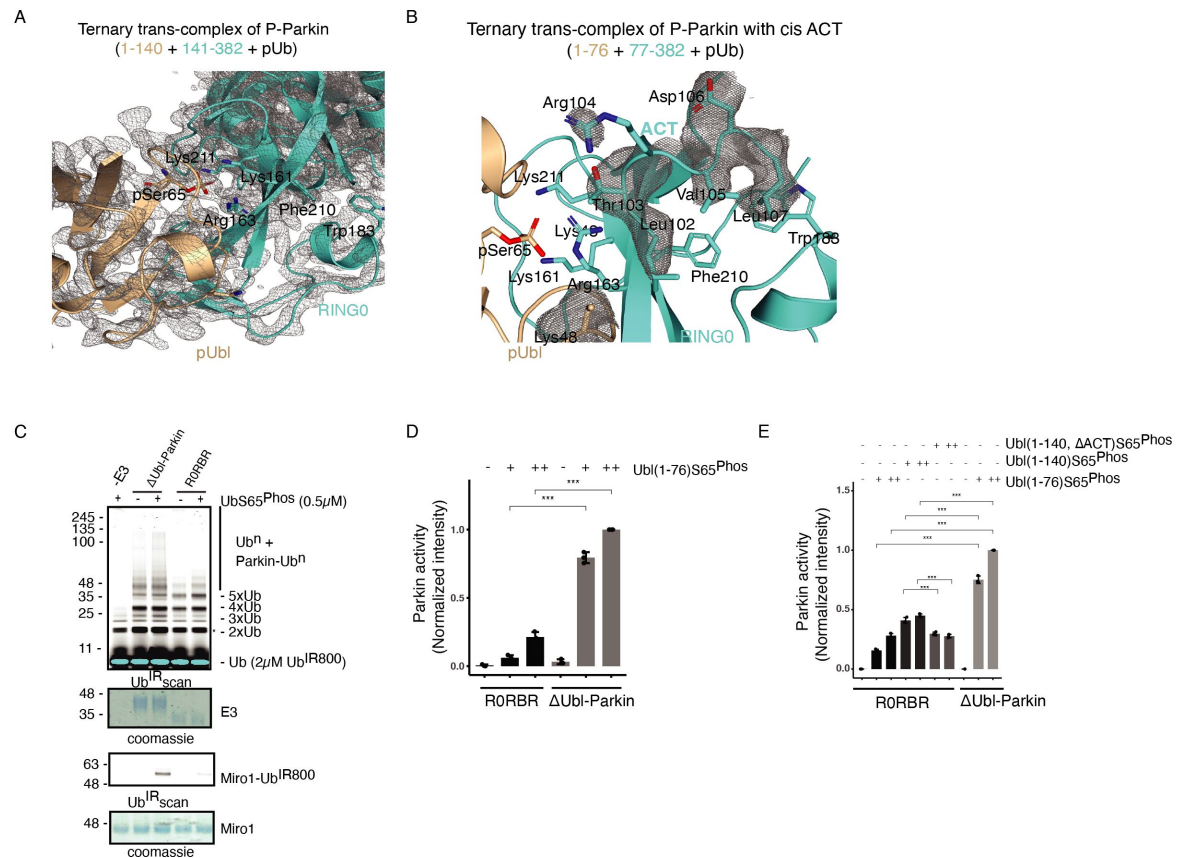
A, B, C, Schematic representation of SEC assay from **Fig 3 A** and **B**, respectively. D. Phos-Tag analysis shows the effect of pUbl (1-76) or pUb (1-76) on the phosphorylation of Parkin by PINK1.



Extended Data Figure 8

Parkin localization on mitochondria.

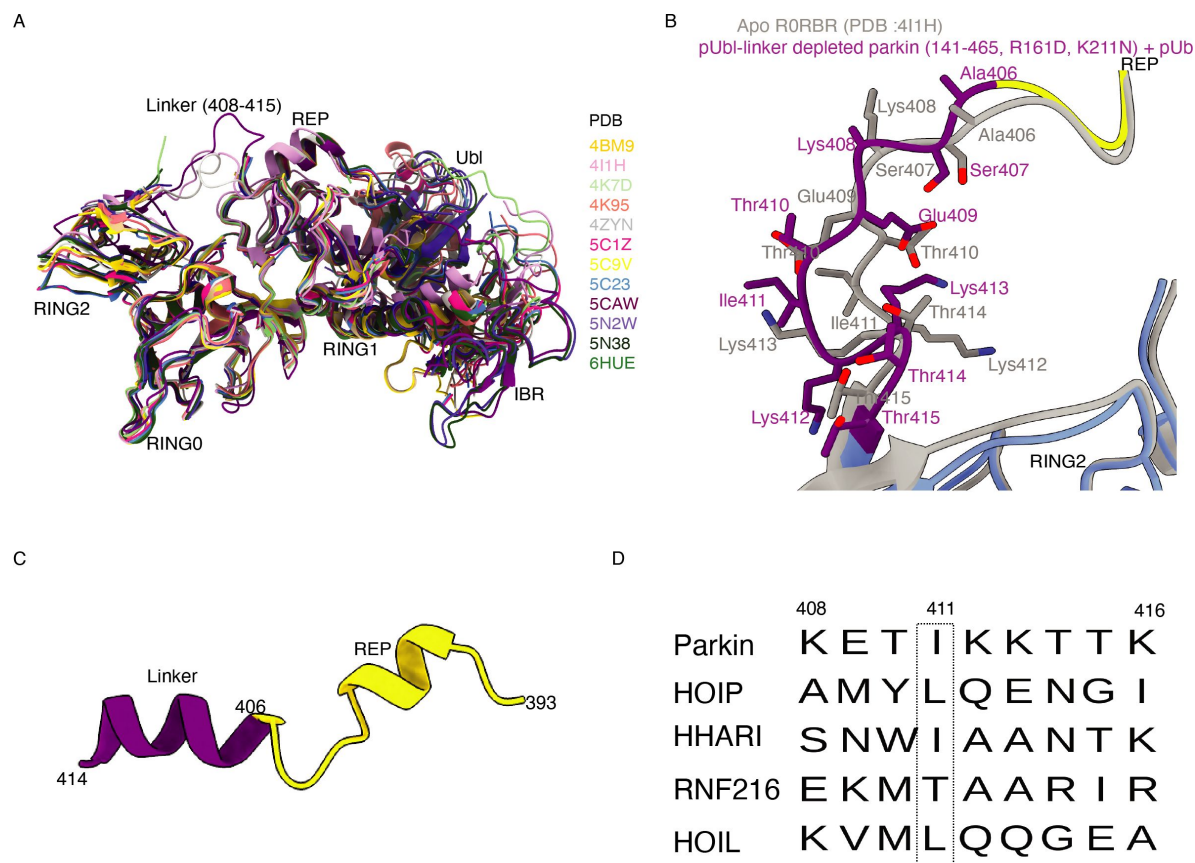
A. HeLa cells were transfected with plasmid for wild-type (WT) GFP-Parkin, GFP-Parkin C431F, GFP-Parkin H302A/C431F and GFP-Parkin K211N/C431F. Cells were treated for 1 h with 10 μ M CCCP or DMSO was used as control. Mitochondria were labeled with anti-TOMM20 antibody (blue). Scale bars, 10 μ m. B. Quantification of GFP-Parkin (WT and mutants) on mitochondria***, $p < 0.001$. C. Quantification of mCherry-Parkin (WT and S65A) on mitochondria in [Figure 4](#), ns=not significant.



Extended Data Figure 9

Role of ACT in Parkin activation.

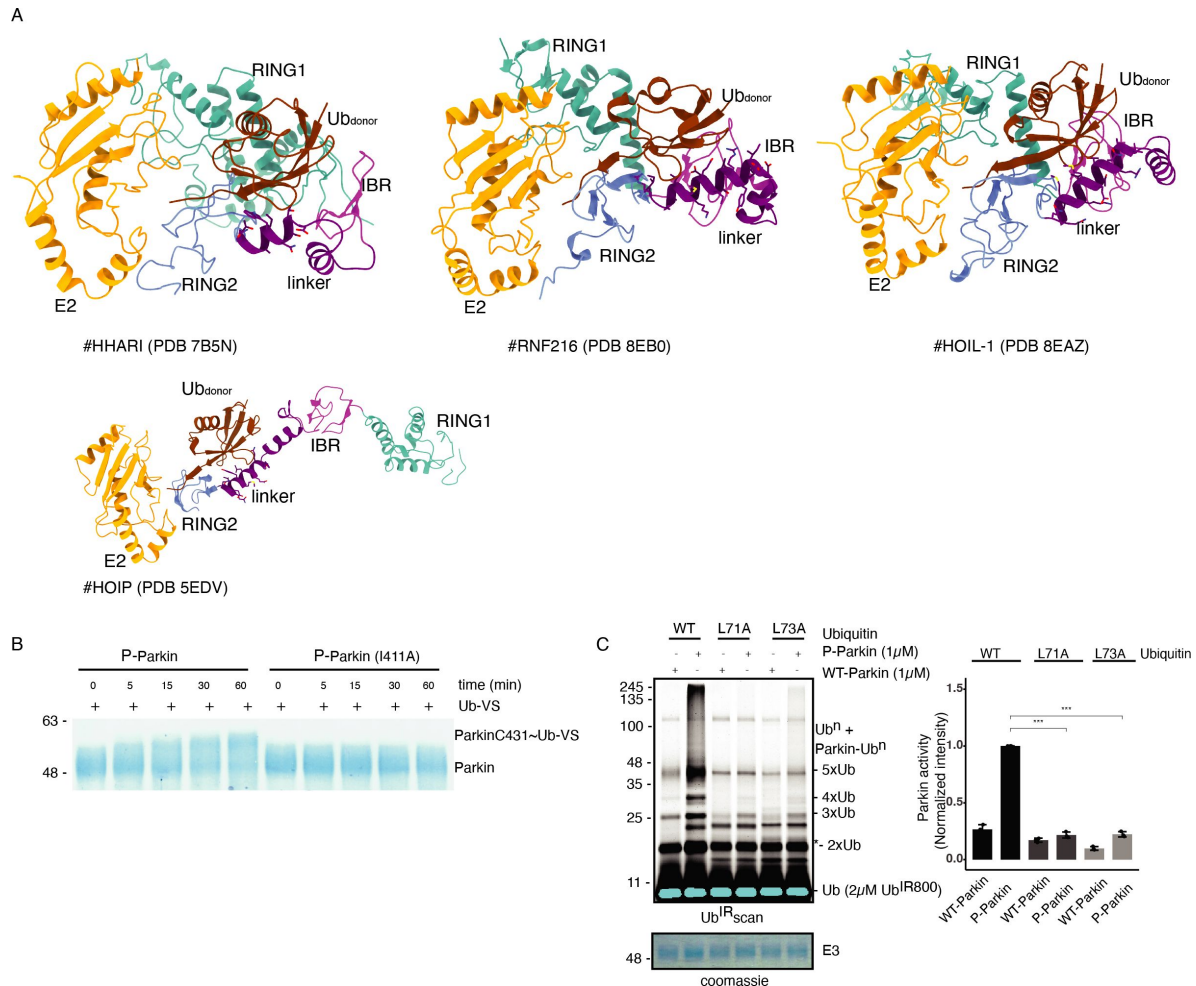
A. The 2Fo-Fc map (grey) of the trans-complex of phospho-Parkin in the crystal structure of the ternary trans-complex of phospho-Parkin (1-140 (wheat) + 141-382 (cyan)). The 2Fo-Fc map is contoured at 1.0 σ . B. The 2Fo-Fc map (grey) of the ACT region in the ternary trans-complex structure of phospho-Parkin with cis ACT (1-76 (wheat) + 77-382 (cyan)). The 2Fo-Fc map is contoured at 1.0 σ . C. Comparison of ubiquitination activity of Δ Ubl-Parkin and R0RBR. A non-specific, ATP-independent band is indicated (*). The middle panel shows a Coomassie-stained loading control. The lower panel shows Miro1 ubiquitination for the respective proteins shown in the upper lane. Coomassie-stained gel showing Miro1 is used as the loading control of substrate ubiquitination assay. D. The bar graph shows the integrated intensities of Miro-1 ubiquitination levels from three independent experiments (mean $\pm$ s.e.m.). pUbl was added in 2-fold (+) or 4-fold (++) molar excess. Statistical significance was determined using pair-wise student's t-test ($P < 0.001$). E. The bar graph shows the integrated intensities of Miro-1 ubiquitination levels from three independent experiments (mean $\pm$ s.e.m.). Various pUbl constructs were added in 2-fold (+) or 4-fold (++) molar excess. Statistical significance was determined using pair-wise student's t-test ($P < 0.001$).



Extended Data Figure 10

The linker connecting REP and RING2 shows conformational flexibility.

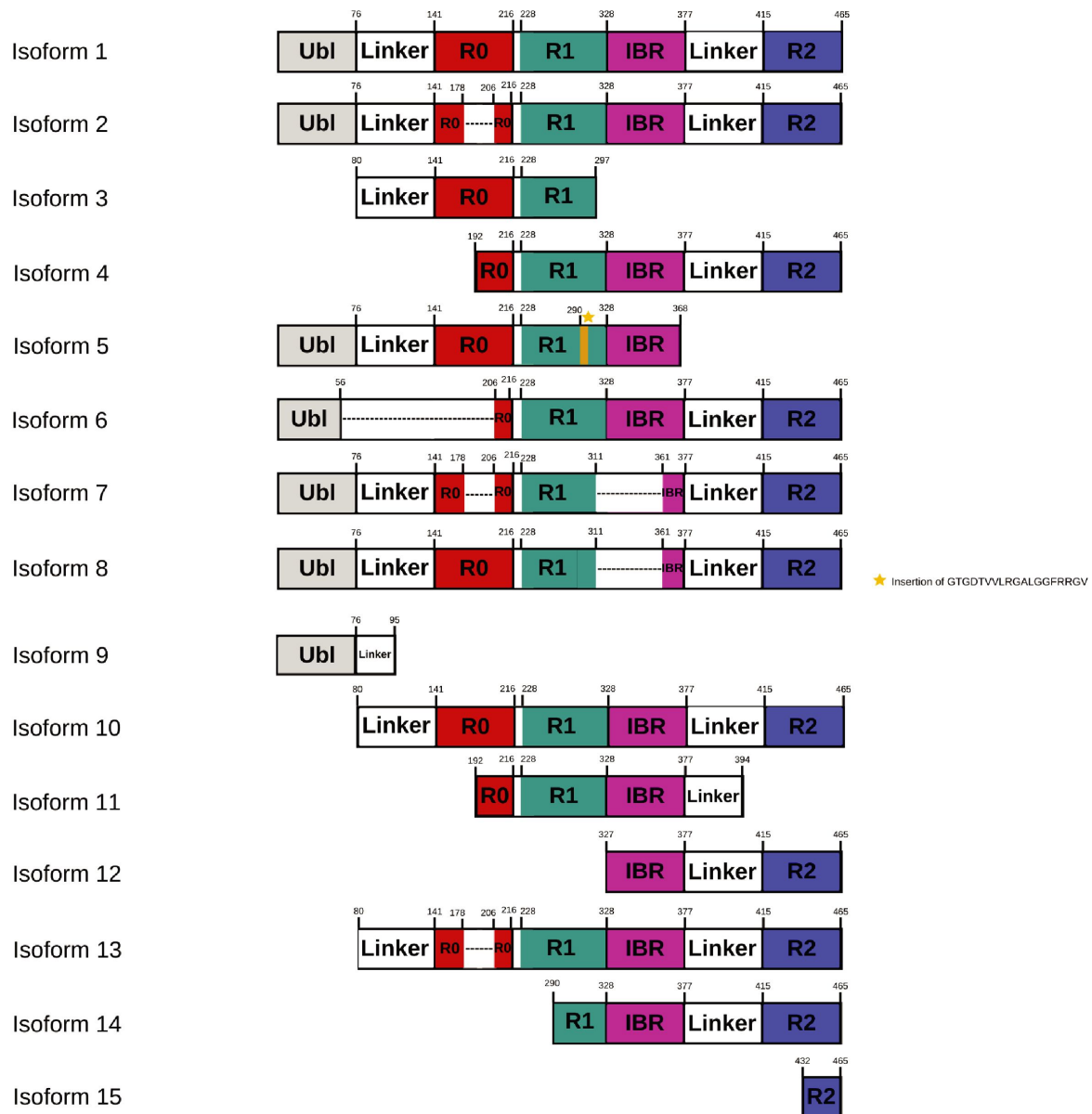
A. Superimposition of Parkin structures showing the flexible nature of the linker between REP-RING2. Different Parkin structures are colored according to their PDB code. B. Superimposition of pUbl-linker (1-140) depleted Parkin (141-465, Arg163Asp, Lys211Asn) and pUb complex with apo R0RBR (144-465, PDB 4I1H) showing conformational changes in the linker region (408-415, purple) of Parkin. Apo R0RBR (PDB 4I1H) structure is shown in grey. C. Conformation of REP-linker predicted from AlfaFold ³⁴. A Full-length Parkin sequence was used to model the Parkin structure using AlfaFold. D. Sequence alignment of the linker connecting IBR-RING2 among various RBR family E3-ligases. The conserved hydrophobic patch among Parkin, HOIP, HHARI, and HOIL is highlighted with a dashed box. The sequence numbering is according to the sequence of Parkin.



Extended Data Figure 11

The linker connecting REP and RING2 domain binds with ubiquitin (Ub_{don}) of E2-Ub.

A. Crystal structures of various RBR family E3-ligases showing linker between IBR-RING2 interact with ubiquitin (Ub_{don}) of E2-Ub. Only RING1 (cyan), IBR (magenta), linker (purple), and RING2 (blue) are shown interacting with E2 (orange) and ubiquitin (Ub_{don}) (brown) of E2-Ub. B. Ub-VS probe reactivity with the catalytic Cys of RING2 of phospho-Parkin, or phospho-Parkin I411A. C. Ubiquitination assay to compare Parkin activity using WT/mutant ubiquitin. A non-specific, ATP-independent band is indicated (). The lower panel shows a Coomassie-stained loading control. The bar graph shows the integrated intensities of ubiquitination levels from three independent experiments (mean ± s.e.m.). Statistical significance was determined using pairwise student's t-test (***P* < 0.001)



Extended Data Figure 12

Schematic representation of domain organization in various isoforms ⁴¹[41](#), ⁴²[42](#) of Parkin.

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Reviewer #1 (Public Review):

Summary:

The authors used structural and biophysical methods to provide insight into Parkin regulation. The breadth of data supporting their findings was impressive and generally well-orchestrated. Still, the impact of their results builds on recent structural studies and the stated impact is based on these prior works.

Strengths:

(1) After reading through the paper, the major findings are:

- RING2 and pUbl compete for binding to RING0.
- Parkin can dimerize.
- ACT plays an important role in enzyme kinetics.

(2) The use of molecular scissors in their construct represents a creative approach to examining inter-domain interactions.

(3) From my assessment, the experiments are well-conceived and executed.

Weaknesses:

(1) The manuscript, as written, is NOT for a general audience. Admittedly, I am not an expert on Parkin structure and function, but I had to do a lot of homework to try to understand the underlying rationale and impact. This reflects, I think, that the work generally represents an incremental advance on recent structural findings.

(2) To this point, it is hard to understand the impact of this work without more information highlighting the novelty. There are several structures of Parkin in various auto-inhibited states, and it was hard to delineate how this is different.

(3) As noted, I appreciated the use of protease sites in the fusion protein construct. It is unclear how the loop region might affect the protein structure and function. The authors worked to demonstrate that this did not introduce artifacts, but the biological context is missing.

(4) While it is likely that the binding is competitive between the Ubl and RING2 domains, the data is not quantitative. Is it known whether the folding of the distinct domains is independent? Or are there interactions that alter folding? It seems plausible that conformational rearrangements may invoke an orientation of domains that would be incompatible. The biological context for the importance of this interaction was not clear to me.

(5) What is the rationale for mutating Lys211 to Asn? Were other mutations tried? Glu? Ala? Just missing the rationale. I think this may have been identified previously in the field, but not clear what this mutation represents biologically.

(6) I was confused about how the phospho-proteins were generated. After looking through the methods, there appear to be phosphorylation experiments, but it is unclear what the efficiency was for each protein (i.e. what % gets modified). In the text, the authors refer to phospho-Parkin (T270R, C431A), but not clear how these mutations might influence this process. I gather that these are catalytically inactive, but it is unclear to me how this is catalyzing the ubiquitination in the assay.

(7) The authors note that "ACT can be complemented in trans; however, it is more efficient in cis", but it is unclear whether both would be important or if the favored interaction is dominant in a biological context.

(8) The authors repeatedly note that this study could aid in the development of small-molecule regulators against Parkin to treat PD, but this is a long way off. And it is not clear

from their manuscript how this would be achieved. As stated, this is conjecture.

<https://doi.org/10.7554/eLife.96699.1.sa2>

Reviewer #2 (Public Review):

This manuscript uses biochemistry and X-ray crystallography to further probe the molecular mechanism of Parkin regulation and activation. Using a construct that incorporates cleavage sites between different Parkin domains to increase the local concentration of specific domains (i.e., molecular scissors), the authors suggest that competitive binding between the p-Ubl and RING2 domains for the RING0 domain regulates Parkin activity. Further, they demonstrate that this competition can occur in trans, with a p-Ubl domain of one Parkin molecule binding the RING0 domain of a second monomer, thus activating the catalytic RING1 domain. In addition, they suggest that the ACT domain can similarly bind and activate Parkin in trans, albeit at a lower efficiency than that observed for p-Ubl. The authors also suggest from crystal structure analysis and some biochemical experiments that the linker region between RING2 and repressor elements interacts with the donor ubiquitin to enhance Parkin activity.

Ultimately this manuscript challenges previous work suggesting that the p-Ubl domain does not bind to the Parkin core in the mechanism of Parkin activation. The use of the 'molecular scissors' approach to probe these effects is an interesting approach to probe this type of competitive binding. However, there are issues with the experimental approach manuscript that detract from the overall quality and potential impact of the work.

The competitive binding between p-Ubl and RING2 domains for the Parkin core could have been better defined using biophysical and biochemical approaches that explicitly define the relative affinities that dictate these interactions. A better understanding of these affinities could provide more insight into the relative bindings of these domains, especially as it relates to the in trans interactions.

I also have concerns about the results of using molecular scissors to 'increase local concentrations' and allow for binding to be observed. These experiments are done primarily using proteolytic cleavage of different domains followed by size exclusion chromatography. ITC experiments suggest that the binding constants for these interactions are in the μM range, although these experiments are problematic as the authors indicate in the text that protein precipitation was observed during these experiments. This type of binding could easily be measured in other assays. My issue relates to the ability of a protein complex (comprising the core and cleaved domains) with a K_d of 1 μM to be maintained in an SEC experiment. The off-rates for these complexes must be exceeding slow, which doesn't really correspond to the low μM binding constants discussed in the text. How do the authors explain this? What is driving the K_{off} to levels sufficiently slow to prevent dissociation by SEC? Considering that the authors are challenging previous work describing the lack of binding between the p-Ubl domain and the core, these issues should be better resolved in this current manuscript. Further, it's important to have a more detailed understanding of relative affinities when considering the functional implications of this competition in the context of full-length Parkin. Similar comments could be made about the ACT experiments described in the text.

Ultimately, this work does suggest additional insights into the mechanism of Parkin activation that could contribute to the field. There is a lot of information included in this manuscript, giving it breadth, albeit at the cost of depth for the study of specific interactions. Further, I felt that the authors oversold some of their data in the text, and I'd recommend being a bit more careful when claiming an experiment 'confirms' a specific model. In many cases, there are other models that could explain similar results. For example, in Figure 1C, the authors state that their crystal structure 'confirms' that "RING2 is transiently displaced from the

RING0 domain and returns to its original position after washing off the p-Ubl linker". However, it isn't clear to me that RING2 ever dissociated when prepared this way. While there are issues with the work that I feel should be further addressed with additional experiments, there are interesting mechanistic details suggested by this work that could improve our understanding of Parkin activation. However, the full impact of this work won't be fully appreciated until there is a more thorough understanding of the regulation and competitive binding between p-Ubl and RING2 to RORB both in cis and in trans.

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Reviewer #3 (Public Review):

Summary:

In their manuscript "Additional feedforward mechanism of Parkin activation via binding of phospho-UBL and RING0 in trans", Lenka et al present data that could suggest an "in trans" model of Parkin ubiquitination activity. Parkin is an intensely studied E3 ligase implicated in mitophagy, whereby missense mutations to the PARK2 gene are known to cause autosomal recessive juvenile parkinsonism. From a mechanistic point of view, Parkin is extremely complex. Its activity is tightly controlled by several modes of auto-inhibition that must be released by queues of mitochondrial damage. While the general overview of Parkin activation has been mapped out in recent years, several details have remained murky. In particular, whether Parkin dimerizes as part of its feed-forward signaling mechanism, and whether said dimerization can facilitate ligase activation, has remained unclear. Here, Lenka et al. use various truncation mutants of Parkin in an attempt to understand the likelihood of dimerization (in support of an "in trans" model for catalysis).

Strengths:

The results are bolstered by several distinct approaches including analytical SEC with cleavable Parkin constructs, ITC interaction studies, ubiquitination assays, protein crystallography, and cellular localization studies.

Weaknesses:

As presented, however, the storyline is very confusing to follow and several lines of experimentation felt like distractions from the primary message. Furthermore, many experiments could only indirectly support the author's conclusions, and therefore the final picture of what new features can be firmly added to the model of Parkin activation and function is unclear.

Major concerns:

(1) This manuscript solves numerous crystal structures of various Parkin components to help support their idea of in trans transfer. The way these structures are presented more resemble models and it is unclear from the figures that these are new complexes solved in this work, and what new insights can be gleaned from them.

(2) There are no experiments that definitively show the in trans activation of Parkin. The binding experiments and size exclusion chromatography are a good start, but the way these experiments are performed, they'd be better suited as support for a stronger experiment showing Parkin dimerization. In addition, the rationale for an in trans activation model is not convincingly explained until the concept of Parkin isoforms is introduced in the Discussion. The authors should consider expanding this concept into other parts of the manuscript.

2a. For the in trans activation experiment using wt Parkin and pParkin (T270R/C431A) (Figure 3D), there needs to be a large excess of pParkin to stimulate the catalytic activity of wt Parkin. This experiment has low cellular relevance as these point mutations are unlikely to occur together to create this nonfunctional pParkin protein. In the case of pParkin activating wt Parkin (regardless of artificial point mutations inserted to study specifically the in trans activation), if there needs to be much more pParkin around to fully activate wt Parkin, isn't it just more likely that the pParkin would activate in cis?

2ai. Another underlying issue with this experiment is that the authors do not consider the possibility that the increased activity observed is a result of increased "substrate" for auto-ubiquitination, as opposed to any role in catalytic activation. Have the authors considered looking at Miro as a substrate in order to control for this?

2b. The authors mention a "higher net concentration" of the "fused domains" with RING0, and use this to justify artificially cleaving the Ubl or RING2 domains from the Parkin core. This fact should be moot. In cells, it is expected there will only be a 1:1 ratio of the Parkin core with the Ubl or RING2 domains. To date, there is no evidence suggesting multiple pUbls or multiple RING2s can bind the RING0 binding site. In fact, the authors here even show that either the RING2 or pUbl needs to be displaced to permit the binding of the other domain. That being said, there would be no "higher net concentration" because there would always be the same molar equivalents of Ubl, RING2, and the Parkin core.

2c. A larger issue remaining in terms of Parkin activation is the lack of clarity surrounding the role of the linker (77-140); particularly whether its primary role is to tether the Ubl to the cis Parkin molecule versus a role in permitting distal interactions to a trans molecule. The way the authors have conducted the experiments presented in Figure 2 limits the possible interactions that the activated pUbl could have by (a) ablating the binding site in the cis molecule with the K211N mutation; (b) further blocking the binding site in the cis molecule by keeping the RING2 domain intact. These restrictions to the cis parkin molecule effectively force the pUbl to bind in trans. A competition experiment to demonstrate the likelihood of cis or trans activation in direct comparison with each other would provide stronger evidence for trans activation.

(3) A major limitation of this study is that the authors interpret structural flexibility from experiments that do not report directly on flexibility. The analytical SEC experiments report on binding affinity and more specifically off-rates. By removing the interdomain linkages, the accompanying on-rate would be drastically impacted, and thus the observations are disconnected from a native scenario. Likewise, observations from protein crystallography can be consistent with flexibility, but certainly should not be directly interpreted in this manner. Rigorous determination of linker and/or domain flexibility would require alternative methods that measure this directly.

(4) The analysis of the ACT element comes across as incomplete. The authors make a point of a competing interaction with Lys48 of the Ubl domain, but the significance of this is unclear. It is possible that this observation could be an overinterpretation of the crystal structures. Additionally, the rationale for why the ACT element should or shouldn't contribute to in trans activation of different Parkin constructs is not clear. Lastly, the conclusion that this work explains the evolutionary nature of this element in chordates is highly overstated.

(5) The analysis of the REP linker element also seems incomplete. The authors identify contacts to a neighboring pUb molecule in their crystal structure, but the connection between this interface (which could be a crystallization artifact) and their biochemical activity data is not straightforward. The analysis of flexibility within this region using crystallographic and AlphaFold modeling observations is very indirect. The authors also draw parallels with linker regions in other RBR ligases that are involved in recognizing the E2-loaded Ub. Firstly, it is not

clear from the text or figures whether the "conserved" hydrophobic within the linker region is involved in these alternative Ub interfaces. And secondly, the authors appear to jump to the conclusion that the Parkin linker region also binds an E2-loaded Ub, even though their original observation from the crystal structure seems inconsistent with this. The entire analysis feels very preliminary and also comes across as tangential to the primary storyline of in trans Parkin activation.

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